Spaceborne SAR Time-Series Radiometric Cross-Calibration via Spatiotemporal Screening-Regression-Modeling Optimization

Yongsheng Zhou, *Member, IEEE*, Jing Su, *Student Member, IEEE*, Lingling Ma, *Member, IEEE*, Qiang Yin, *Senior Member, IEEE*, and Fan Zhang, *Senior Member, IEEE*

***Abstract*—Maintaining temporal radiometric consistency in synthetic aperture radar (SAR) satellite imagery requires advanced time-series radiometric calibration. Conventional approaches that rely on artificial calibration targets incur high operational costs. Capitalizing on the cost-efficiency of natural-target-based radiometric cross-calibration, this study extends cross-calibration principles to time-series calibration through three innovations: 1) a multicriterion calibration target screening method through integrated analysis combining satellite revisit frequencies, target spatial histogram characteristics, and temporal scattering stability to overcome spatiotemporal constraints; 2) an enhanced calibration constant calculation approach that iteratively applies robust linear regression and removal of spatial pixel outliers to improve radiometric accuracy; 3) an intertemporal calibration constant optimization mechanism is implemented by implementing Kalman filtering and sensor-attenuation modeling to ensure long-term stability. Experimental validation using Sentinel-1 data reveals that the maximum root-mean-square error (RMSE) across 69 temporal points is 0.36 dB, representing a 62% reduction compared to conventional methods. Furthermore, the Kalman-filtered results achieve a temporal stability of 0.0269 dB, an 89% improvement over unprocessed calibrations. This methodology enables continuous monitoring of SAR sensor performance with improved temporal consistency, which is crucial for the long-term harmonization of radiometric data.**

***Index Terms*—Cross-calibration, Kalman filter, radiometric calibration, robust regression, synthetic aperture radar (SAR), time-series calibration.**

Received 28 February 2025; revised 28 August 2025, 2 December 2025, and 31 December 2025; accepted 4 January 2026. Date of publication 12 January 2026; date of current version 27 February 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62171016 and Grant 62271034 and in part by the Fundamental Research Funds for the Central Universities under Grant buctrc202001. *(Corresponding author: Lingling Ma.)*

Yongsheng Zhou, Jing Su, Qiang Yin, and Fan Zhang are with the College of Information Science and Technology, Beijing University of Chemical Technology, Beijing 100029, China (e-mail: zhyosh@mail.buct.edu.cn; 2022200839@buct.edu.cn; yinq@mail.buct.edu.cn; zhangf@mail.buct.edu.cn).

Lingling Ma is with the National Engineering Laboratory for Satellite Remote Sensing Applications, Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100094, China (e-mail: mall@aircas.ac.cn).

Digital Object Identifier 10.1109/TGRS.2026.3652214

I. Introduction

SYNTHETIC aperture radar (SAR) has emerged as a pivotal Earth observation technology, providing high-resolution microwave images through platform motion synthesis and coherent signal processing [1]. Unlike passive optical systems constrained by solar illumination and atmospheric conditions, SAR offers unique all-weather and day–night imaging capabilities that have enabled critical advancements in topographic mapping [2], environmental monitoring [3], [4], and disaster response [5].

Radiometric calibration systematically constructs a dimensionally consistent mathematical framework that bridges raw digital number (DN) values (typically unitless integers ranging from 0 to 65 535) to scientifically interpretable geophysical parameters, most notably the backscattering coefficient (σ^0) with the unit of $\mathrm{dBm^2/m^2}$, as illustrated in Fig. 1. This process compensates for systematic errors from antenna pattern variations, system gain fluctuations, and thermal noise [6]. The derived radiometric calibration constant is a key parameter for processing SAR data products from low (L1A) to advanced level (L1B). Accurate radiometric calibration is indispensable for quantitative analysis, such as biomass estimation [7] and soil moisture inversion [8].

However, because of factors such as component aging and environmental variations, the radiometric performance of sensors in orbit inevitably drifts. This drift causes deviations in the calibration results, which affects the accurate representation of ground object characteristics. To address this issue, time-series radiometric calibration extends conventional single-temporal approaches by periodically updating calibration constants to monitor sensor drift and ensure temporal consistency. This capability is vital for long-term monitoring applications [9]. In the domain of optical sensor calibration, time-series radiometric calibration has been extensively applied, mainly via field calibration and cross-calibration methods. It was initially implemented on the National Oceanic and Atmospheric Administration/Advanced Very High Resolution Radiometer (NOAA/AVHRR) series of meteorological satellites. Nagaraja Rao and Chen [10], Cosnefroy et al. [11], and Heidinger et al. [12] used multiple desert areas in Africa as

1558-0644 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:27:37 UTC from IEEE Xplore. Restrictions apply.

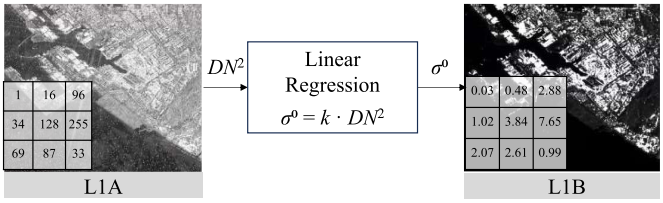


Fig. 1. Illustration of SAR radiometric calibration.

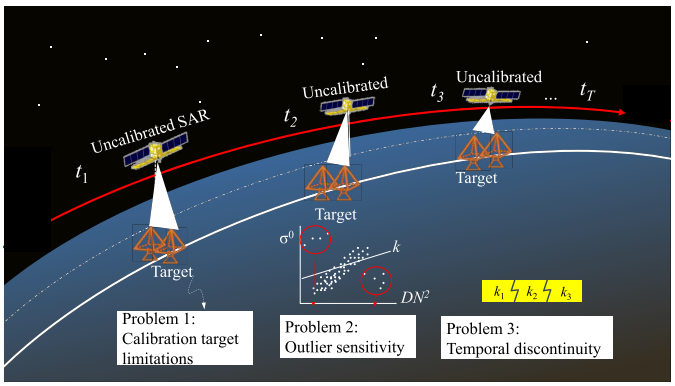


Fig. 2. Illustration of the existing time-series calibration methods and the problems.

reference targets to obtain time-series calibration constants for the entire life cycle of AVHRR. Subsequently, Wei et al. [13] proposed a multisite calibration method to obtain time-series calibration constants for GF-1/WFV, demonstrating good consistency with the official calibration results. Meanwhile, the radiometric cross-calibration method has also been widely utilized in optical sensor calibration. Cabot et al. [14] used North Africa and Saudi Arabia as calibration targets and the Polarization and Directionality of the Earth's Reflectances (POLDERs) sensor as a reference benchmark, achieving time-series calibration for Vegetation, AVHRR, and the sea-viewing wide field-of-view sensor (SeaWiFS). Wang et al. [15] and Liu et al. [16], respectively, used Sentinel-2/Multispectral Instrument (Sentinel-2/MSI) and Terra/Moderate Resolution Imaging Spectroradiometer (Terra/MODIS) as reference satellites, and the radiometric calibration network (RadCalNet) sites and the Dunhuang site as calibration targets, achieving long-term radiometric cross-calibration for GF-6/WFV and HJ-1A/CCD. It is worth mentioning that Dou et al. [17] proposed a Kalman-filter-based time-series calibration method, which achieved certain optimization effects. However, the process neglected the critical impact of initial parameter settings on the filter, leading to suboptimal filtering or even divergence.

In contrast, SAR time-series calibration methodologies remain comparatively underdeveloped. Current operational systems such as Sentinel-1A/1B and TerraSAR-X/TanDEM-X primarily utilize ground reference target monitoring to track radiometric performance [18], [19]. Although these implementations successfully capture basic drift patterns, their fundamental constraints become apparent when compared with optical calibration methods, as illustrated in Fig. 2, and there are still some critical limitations as follows.

1) *Calibration Target Limitations:* The current SAR time-series calibration methods are mainly based on artificial calibration targets (corner reflectors and transponders) or rainforests [20], [21], [22], as illustrated in Fig. 2. These artificial calibration targets face three fundamental constraints: 1) the development and deployment costs are extremely high, making them difficult to scale for large-area monitoring; 2) they are susceptible to environmental erosion and human destruction, resulting in insufficient long-term stability; and 3) their geographical distribution is limited, making frequent implementation difficult. Chen et al. [23] proposed a method to extract pseudo-invariant distributed targets in 2023 and established a global database, including urban areas, saline-alkali lands, and rainforests. Shi et al. [24] and Shangguan et al. [25] used rainforest and urban areas to monitor the radiometric stability of LT-1A and Sentinel-1, respectively. Zhou et al. [26], and Deng et al. [27], respectively, used saline-alkali lands and urban areas as reference targets to achieve SAR radiometric cross-calibration. However, existing target screening methods are poorly developed for heterogeneous targets such as urban areas and rarely consider the imaging overlap frequency. To address this, we introduce a multicriterion method integrating satellite revisit frequencies, spatial histogram characteristics, and temporal scattering stability.
2) *Sensitivity of Calibration Constant Calculation to Data Outliers:* SAR radiometric cross-calibration constants are typically derived by linear regression between the DN values of the calibration targets and their corresponding backscattering coefficients (σ^0) [28]. In the domain of optical sensor calibration, in 2020, Gao et al. [29] calculated weights based on the quality differences of the data pairs and obtained the radiometric cross-calibration constant using a weighted regression method. In 2024, Zhou et al. [30], [31] introduced the weighted regression method into SAR radiometric cross-calibration. The weights were designed based on the uncertainties of distributed target blocks and the quality differences of the target sites, and the radiometric calibration constant was then calculated. The results indicate that both the weighting of the target blocks and the weighting of different calibration sites effectively improve the precision of the radiometric calibration. However, existing methods only considered the uncertainty of the data blocks. They did not account for outlier interference in the extracted data points, which will affect the precision of the radiometric calibration. This study presents an enhanced approach that iteratively applies robust linear regression and removal of spatial pixel outliers, mitigating the negative impact of outliers on calculating the calibration constants.
3) *Neglect of the Temporal Correlation of the Calibration Constants:* Most time-series radiometric cross-calibration methods obtain calibration constants individually without considering the relationship between calibration constants at different temporal points [32],

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:27:37 UTC from IEEE Xplore. Restrictions apply.

[33]. In reality, the in-orbit state of the remote sensor is a continuous and gradually changing process over time [34]. There should be a specific correlation between its calibration constants. Dou et al. [17] attempted to introduce the Kalman filter into the time-series calibration of optical sensors, using the calibration result of the previous temporal point as the predicted value to optimize the calibration constant at the current temporal point. The effectiveness of optimization is closely related to the initial settings of the prediction error and measurement error, but that study did not involve the criteria for establishing these values. In response to the above issues, this study proposes an intertemporal calibration constant optimization mechanism by implementing the Kalman filter. Based on the radiometric stability of the SAR sensor and the nominal calibration accuracy, the initial values of the prediction error and measurement error are reasonably set to optimize the time-series calibration constants and effectively reduce the uncertainty of the single-temporal radiometric calibration constants.

In summary, the main contributions of this article are as follows.

1) An SAR time-series radiometric cross-calibration method using naturally distributed targets is proposed. Unlike artificial calibration target-based approaches, it ensures high accuracy while meeting high-frequency calibration requirements.
2) A multicriterion target screening method is developed, integrating satellite revisit frequency, spatial histogram characteristics, and temporal stability. This approach improves target selection in heterogeneous areas (e.g., urban regions) over traditional single-criterion methods.
3) An iterative robust regression method with outlier removal is proposed to estimate single-temporal calibration constants. This method effectively mitigates outlier interference, producing robustness higher than that of conventional regression methods.
4) An intertemporal optimization mechanism using the Kalman filter is introduced. By modeling correlations between temporal points, it reduces uncertainty in time-series results and better captures sensor drift compared to existing methods.

The remainder of this article is structured as follows. Section II details the proposed method. In Sections III and IV, the experimental results are presented and discussed, respectively. Section V concludes the article.

II. Methodology

A. Overall Calibration Process

The proposed method establishes a systematic framework for SAR time-series radiometric cross-calibration (Fig. 3). It begins with screening the calibration target areas based on three crucial criteria: satellite revisit frequencies, target spatial histogram characteristics, and temporal scattering stability. Subsequently, calibrated and uncalibrated SAR image pairs for radiometric cross-calibration are obtained by precisely matching the geolocations of the identified regions. Then, an iterative robust regression method with outlier removal (IRROR) is used to calculate the calibration constant at each temporal point. The temporal calibration consistency is further strengthened using the Kalman filter, which concurrently facilitates the construction of sensor degradation models by analyzing the residual trends in time-series calibration constants.

B. Step 1: Screening of the Calibration Target Area and Acquisition of Time-Series SAR Image Pairs

The fundamental principle of SAR radiometric cross-calibration lies in the coordinated acquisition of imagery over identical calibration targets by both calibrated and uncalibrated SAR systems. This method essentially facilitates the transfer of the radiometric benchmark from calibrated SAR to uncalibrated SAR through a carefully selected intermediate medium—the calibration reference target.

Natural distributed targets for SAR radiometric cross-calibration can be broadly classified into two categories: homogeneous targets (e.g., saline-alkali lands, bare lands, and rainforests) characterized by spatial uniformity (spatial coefficient of variation <5%) and heterogeneous targets (e.g., urban areas) exhibiting composite scattering mechanisms. Although urban targets exhibit spatial variability, recent studies [35], [36] indicate their remarkable temporal stability when properly screened. Moreover, urban areas provide higher σ^0 values than homogeneous targets in co-polarization modes [37]. The median of σ^0 can reach approximately -6 to -8 dB, providing a higher signal-to-noise ratio for calibration. Their widespread global distribution and frequent satellite coverage further increase their suitability as calibration targets.

Existing screening methods for distributed targets have been well-developed for homogeneous targets. For example, Zhou et al. [31] proposed a pseudo-invariant target screening method based on spatiotemporal characteristics. Based on Sentinel-1 data, the spatial variation coefficient (SVC) is used to construct masks with temporal scattering analysis to screen calibration targets. However, the mask-based method is ineffective for heterogeneous targets, such as urban areas, which lack spatial uniformity. Currently, the screening of urban targets is based on visual interpretation. Moreover, current approaches neglect the critical parameter of the satellite revisit frequency, which is a key determinant of the feasibility of time-series calibration.

To address these limitations, we integrate satellite revisit frequency analysis with target spatiotemporal characteristics. The flowchart is illustrated in Fig. 4.

1) Phase 1: Screening by Satellite Revisit Frequency Analysis: First, we analyze the frequency of acquisitions that overlap over an urban area using the calibrated SAR and the uncalibrated SAR using orbital parameters and historical acquisition records. Furthermore, remote sensing images may have image quality issues that affect frequency judgment [38]. Therefore, we introduce a data quality factor (DQF) into the frequency calculation formula. The DQF range is from 0 to 1. A value of 1 indicates perfect data quality, while values less

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:27:37 UTC from IEEE Xplore. Restrictions apply.

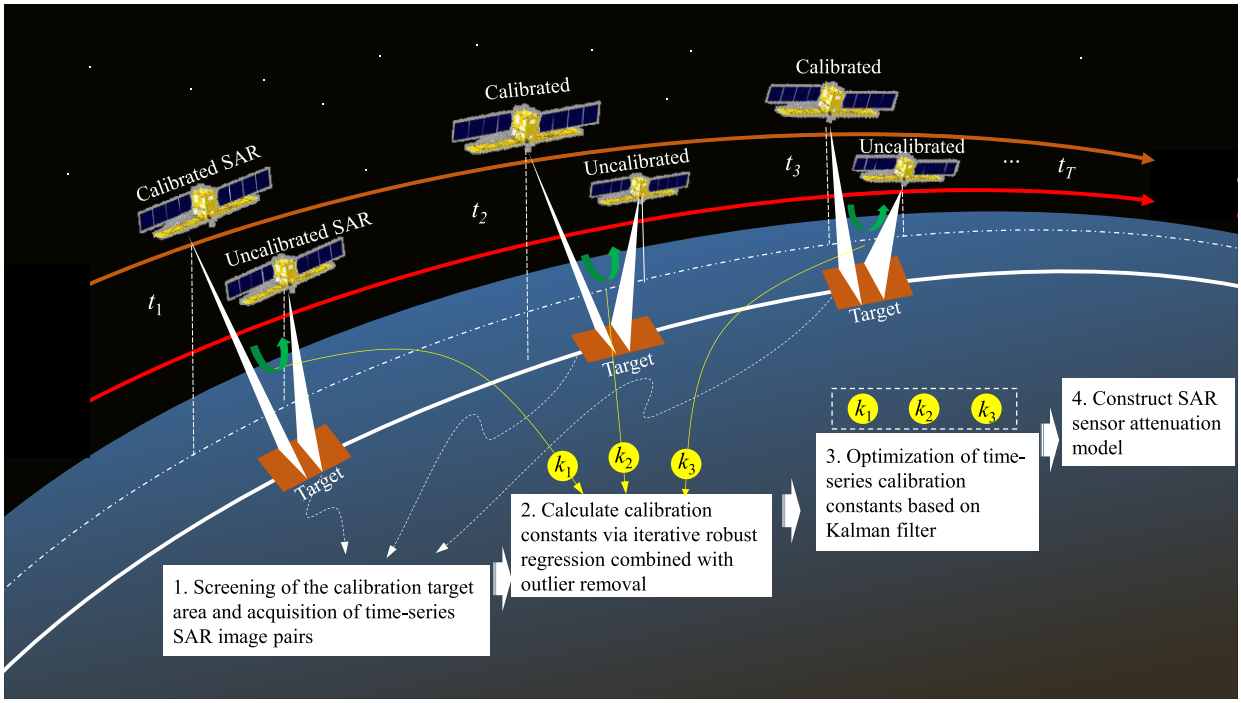


Fig. 3. Flowchart of the method for SAR time-series radiometric cross-calibration.

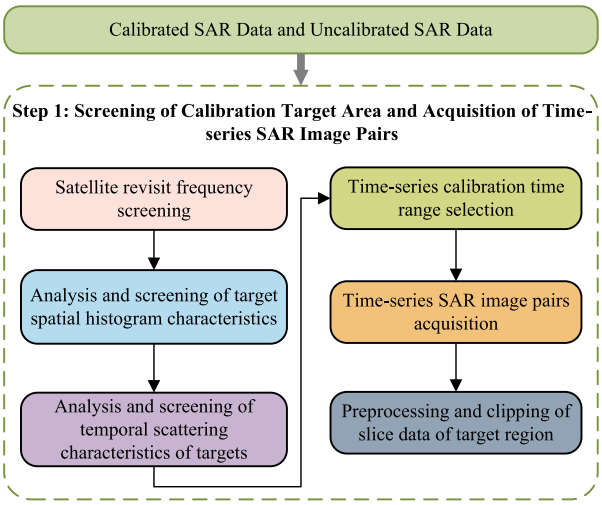


Fig. 4. Flowchart of calibration target area screening and time-series SAR image pairs acquisition.

than 1 indicate a gradual decrease in data quality. Based on the mission duration and revisit cycles, the screening criterion is derived as follows:

$$F_{\text{overlap}} \geq \left\lceil \frac{T_{\text{mission}}}{\max\left(T_{\text{caledSAR}}, T_{\text{uncalSAR}}\right)} \cdot \text{DQF} \right\rceil \tag{1}$$

where F_{overlap} represents the actual number of data pairs available for time-series calibration obtained through statistics, T_{mission} represents the time range for time-series calibration with the unit being days, and T_{caledSAR} and T_{uncalSAR} represent the calibrated and uncalibrated SAR revisit periods, with the unit being days. The right side of the inequality calculates the theoretical minimum number of effective observations. Only target areas where the actual F_{overlap} satisfies this condition are selected. This ensures a sufficient temporal sampling density for time-series analysis and calibration.

2) Phase 2: Screening by Target Spatial Histogram Characteristics: Distributed targets in natural environments typically exhibit contiguous spatial characteristics, forming extensive homogeneous regions. However, only specific subareas within these distributed targets satisfy the essential requirements for radiometric cross-calibration, particularly those demonstrating sufficient homogeneity and temporal stability. Traditional screening methods [23] for such subareas mainly involve the following two screening criteria: SVC and spatial autocorrelation coefficient (SAC). Their joint constraints require target subareas to simultaneously satisfy low spatial variation and high spatial autocorrelation, thereby ensuring spatial consistency in radiometric characteristics. However, urban heterogeneous areas exhibit multiscale structural features: vertical heterogeneity from high-rise buildings and low structures, abrupt reflectivity changes from diverse materials (glass, concrete, and asphalt), and coexistence of multiple land cover types (water bodies, buildings, vegetation, and shadows). These characteristics inherently induce high SVC (due to material reflectivity variations) and low SAC (caused by shadow-induced intensity discontinuities), essentially because this method only captures second-order spatial statistics while failing to characterize the multimodal distribution of complex land cover types.

To address this, we propose a method based on the histogram distribution of backscattering coefficients for screening heterogeneous subareas in urban regions. The histogram of the backscattering coefficients, defined as

$$P_s\left(\sigma_i^0\right) = \frac{M\left(\sigma_i^0\right)}{N} \quad (i = 1, 2, \ldots, j) \tag{2}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:27:37 UTC from IEEE Xplore. Restrictions apply.

where σ_i^0 is the backscattering coefficient value level, $P_s(\sigma_i^0)$ is the probability of this σ_i^0 value occurring, $M(\sigma_i^0)$ is the pixel count at σ_i^0, N is the number of pixels within a subarea, and j is the number of σ_i^0 value levels, intrinsically reflects the statistical distribution of backscattering intensity.

To quantify the histogram morphology, we introduce skewness (β_s) and kurtosis (β_k) as key statistical metrics. Skewness measures the asymmetry of the distribution relative to the mean, while kurtosis describes the peakedness and tail heaviness. They are calculated as follows [39]:

$$\beta_s = \frac{\frac{1}{N}\sum_{m=1}^{N}\left(\sigma_m^0 - \bar{\sigma}^0\right)^3}{\left(\frac{1}{N}\sum_{m=1}^{N}\left(\sigma_m^0 - \bar{\sigma}^0\right)^2\right)^{3/2}} \tag{3}$$

$$\beta_k = \frac{\frac{1}{N}\sum_{m=1}^{N}\left(\sigma_m^0 - \bar{\sigma}^0\right)^4}{\left(\frac{1}{N}\sum_{m=1}^{N}\left(\sigma_m^0 - \bar{\sigma}^0\right)^2\right)^2} - 3 \tag{4}$$

where N is the total number of pixels in the subarea, σ_m^0 is the backscattering coefficient of the mth pixel, and $\bar{\sigma}^0$ denotes the mean backscattering coefficient.

Considering that the skewness and kurtosis of a normal distribution are 0 [40], we select regions with an absolute skewness value less than 0.1 and a kurtosis value less than 0 to meet the following criteria: areas with uniform image brightness, no abnormal bright spots, and a unimodal histogram that is approximately symmetric and has a short-tailed distribution. The unimodality characterizes the uniqueness of the dominant scattering mechanism. The symmetry represents the spatial uniformity of the scattering intensity, and the short-tailed characteristic implies that the disturbance of outliers to the statistical distribution is minimal, meeting the core requirement of cross-calibration for radiometric consistency. From an information theory perspective, the histogram entropy (H_σ) can directly quantify the complexity of the region. A low-entropy state corresponds to a unimodal and short-tailed distribution, satisfying the requirement for information stability [41]

$$H_\sigma\left(\sigma^0\right) = -\sum_{i=1}^{l} P_s\left(\sigma_i^0\right) \cdot \log P_s\left(\sigma_i^0\right) \tag{5}$$

where $P_s(\sigma_i^0)$ is defined in (2), and l is the number of quantization levels.

3) Phase 3: Screening by Target Temporal Scattering Stability: The core principle of radiometric cross-calibration involves establishing a radiometric relationship between two SAR systems through a common reference target. Specifically, the values of the DN acquired by the uncalibrated SAR and the values of backscattering coefficients (σ^0) from the calibrated SAR over the identical target are statistically correlated to calculate the calibration constant. This methodology assumes the stability of the reference target over time.

However, a critical challenge arises from potential temporal variations in the target's scattering properties. The backscattering coefficient is jointly determined by the incidence angle θ, radar wavelength λ, polarization p, soil moisture m_v, surface root mean square height h_{rms}, dielectric constant γ, etc. [42]

$$\sigma^0 = f\left(\theta, \lambda, p, h_{\text{rms}}, m_v, \gamma, \cdots\right) \tag{6}$$

where $f(\cdot)$ denotes a general functional form. If the surface characteristics (e.g., soil moisture and vegetation cover) undergo substantial changes during the intersatellite observation period, the σ^0 values no longer represent the actual scattering state corresponding to the acquisition time of the uncalibrated SAR. Such discrepancies introduce systematic errors into the calibration equation, compromising radiometric cross-calibration accuracy.

To address this, we use a multitemporal stability screening approach based on time-series backscattering analysis. For n coregistered SAR acquisitions spanning a full annual cycle (to account for seasonal variations), we compute the temporal stability metric

$$\text{STD}_\sigma = \sqrt{\frac{1}{n-1}\sum_{t=1}^{n}\left(\sigma_t^0 - \bar{\sigma}_t^0\right)^2} \tag{7}$$

where σ_t^0 represents the spatially median backscattering coefficient at the temporal point t and $\bar{\sigma}_t^0$ denotes the temporal mean over the observation period. After speckle filtering, the spatial median is obtained by extracting the median value of the target area's image. The STD criterion is set at 0.5 dB [31], guaranteeing that the selected subarea maintains radiometric consistency within the radiometric accuracy of modern SAR sensors. If the STD value of a subarea is lower than this value, it is identified as the target area for time-series radiometric cross-calibration in the subsequent procedure, and the latitude and longitude position information is acquired.

4) Phase 4: Acquisition of Time-Series SAR Image Pairs: Finally, calibrated and uncalibrated SAR images for radiometric cross-calibration at each temporal point are screened individually to form calibration target image pairs. The following operations are carried out: 1) following steps such as orbit correction, radiometric calibration, and terrain correction, the backscattering coefficient image of the calibrated SAR and the DN image of the uncalibrated SAR are obtained and 2) the images are cropped to obtain the calibration target image pairs based on the latitude and longitude information of the target area recorded in the above steps.

C. Step 2: Calculation of Single-Temporal Calibration Constant Based on Iterative Robust Regression With Outlier Removal (IRROR)

Reliable time-series radiometric cross-calibration relies on multiple single-temporal radiometric cross-calibrations. Hence, we first improve the accuracy of the single-temporal radiometric cross-calibration method. This section initially presents the radiometric calibration equation, subsequently analyzes the issues of existing methods in solving the radiometric calibration equation, and ultimately presents the solution, namely, the iterative application of robust linear regression and the removal of spatial pixel outliers. Although urban areas may exhibit local variations due to construction activities or seasonal vegetation changes, previous studies [25], [26], [31], [35] have demonstrated that stable scattering points can nevertheless be reliably identified through long-term time-series analysis. For example, Deng et al. [27] demonstrated

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:27:37 UTC from IEEE Xplore. Restrictions apply.

that rooftops, paved surfaces, and other typical urban structures exhibit consistent scattering behavior and can function as reliable radiometric references across multitemporal SAR acquisitions. In this study, the proposed robust regression method builds on these findings by automatically suppressing unstable pixels with abnormal backscattering characteristics, enabling the derived calibration constants to reflect the statistically stable response of persistent urban structures such as roofs and roads.

1) Radiometric Calibration Equation: The SAR radiometric cross-calibration is to establish a calibration equation that relates the DN value of the SAR image with the backscattering coefficients (σ^0) of the calibration target. The radiometric calibration equation is derived from the radar equation, which accounts for the power received by the receiving antenna of the SAR system [43]

$$P_r = \frac{P_t G_t G_r g_t g_r \lambda^2}{(4\pi)^3 R_1^2 R_2^2}\sigma \tag{8}$$

where P_t and P_r are the transmitting and receiving powers of the SAR system. R_1 and R_2 are the distances between the transmitting/receiving antennas and the calibration target. σ is the radar cross section of the calibration target. G_t and G_r are the gain patterns of the transmitting and receiving antennas. g_t and g_r are the gains of the transmitting and receiving antennas, and λ is the radar wavelength.

Generally, the radar system operates by transmitting and receiving radar waves through the same antenna. Therefore, $R_1 = R_2$, $G_t = G_r$, and $g_t = g_r$. Thus, (8) is simplified as

$$P_r = \frac{P_t G^2 g^2 \lambda^2}{(4\pi)^3 R^4}\sigma. \tag{9}$$

Consequently, the radar equation is simplified into a relationship determined by the total transfer function of the SAR system, as follows:

$$P_r = H \cdot \sigma \tag{10}$$

where H is the total transfer function, expressed as

$$H = \frac{P_t G^2 g^2 \lambda^2}{(4\pi)^3 R^4}. \tag{11}$$

In the SAR image, there is the following relationship between the square of the DN value and the receiving power:

$$\mathrm{DN}^2 = K_p \cdot P_r \tag{12}$$

where K_p is the proportional coefficient between DN^2 and the receiving power P_r.

In addition, the relationship between the backscattering coefficient (σ^0) and the radar cross section (σ) is

$$\sigma = \sigma^0 \cdot S_p \tag{13}$$

where S_p is the area of a single pixel in the satellite image.

By combining (10)–(13), the relationship between DN and σ^0 is obtained as

$$\mathrm{DN}^2 = K_p \cdot H \cdot \sigma^0 \cdot S_p. \tag{14}$$

From (14), the radiometric calibration equation in this article is derived as follows:

$$\sigma^0 = k \cdot \mathrm{DN}^2 \tag{15}$$

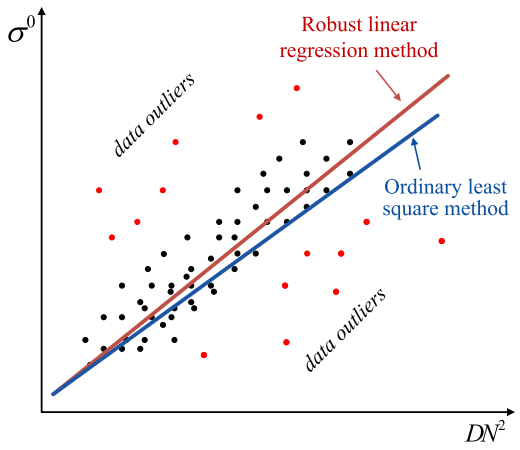


Fig. 5. Comparison of calculation results between the OLSs and robust linear regression methods.

where k is the radiometric calibration constant to be calculated. This equation assumes a zero intercept, as the contribution from system noise is considered negligible for the high-SNR targets used in this study.

2) Existing Problems for Solving the Radiometric Calibration Equation: The traditional radiometric calibration method involves the deployment of artificial calibration targets (such as corner reflectors and transponders) with known σ. The radiometric calibration constant is calculated by combining the DN value of the artificial calibration targets. However, traditional methods that rely on artificial calibration targets face prohibitive costs and a limited deployment density for sustained monitoring.

Given the limitations of traditional methods, natural target-based radiometric cross-calibration methods have emerged as a superior alternative [26]. This method uses widely distributed stable ground targets in nature as alternative reference targets whose backscattering coefficients (σ^0) are obtained by a calibrated SAR. To obtain an accurate σ^0, calibrated and uncalibrated SAR image pairs should be closely matched in time, space, and observation geometry.

The ordinary least squares (OLSs) method is usually used to estimate the parameters (i.e., the radiometric calibration constant k) of a linear model [i.e., the radiometric calibration equation expressed in (15)] by minimizing the sum of the squared differences between the actual value (y) and the predicted value ($\hat{y}$). However, its application requires the following conditions.

1) There is a linear relationship between the independent variable x and the predicted variable $\hat{y}$, which can be expressed as $\hat{y} = \beta_0 + \beta_1 x_1 + \varepsilon$.
2) The residual terms ε are independent of each other, that is, when $i \neq j$, $\mathrm{cov}(\varepsilon_i, \varepsilon_j) = 0$, where $\varepsilon = y - \hat{y}$.
3) The residual term ε must satisfy a normal distribution with a mean of 0 and a variance of ρ^2: $\varepsilon \sim N(0, \rho^2)$.

In reality, even though naturally distributed targets boast high scattering intensity and stability advantages, the complexity of ground features leads to numerous outliers. The presence of outliers violates the assumptions of OLS, such as the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:27:37 UTC from IEEE Xplore. Restrictions apply.

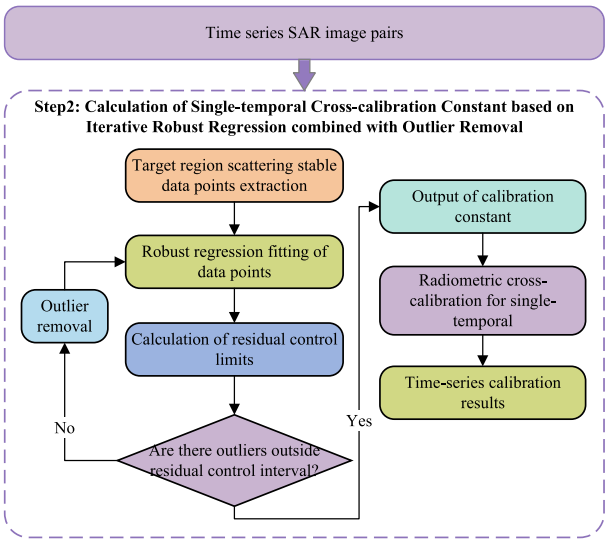


Fig. 6. Flowchart of single-temporal calibration constant calculation based on IRROR.

normality and homoscedasticity of the residuals. The regression results will be biased toward outliers, making the regression line less representative of the overall data pattern (Fig. 5), thus reducing the calculation accuracy of the calibration constant.

3) Enhanced Method for Solving the Radiometric Calibration Equation: To address the issues above, we use the robust regression method, instead of the traditional OLS method, to solve the radiometric calibration equation in (15), which assigns lower weights to residuals that exceed a certain threshold and reduces the influence of outliers. The conventional OLS method treats all pixel data points equally, while robust regression provides more reliable and stable calibration constant results by down-weighting outliers. In addition, although robust regression can reduce the influence of outliers, there may still be some extreme outliers that cannot be eliminated by robust regression. To further refine the data and improve regression accuracy, the residual control interval is calculated to eliminate outliers.

The iterative process of outlier removal and regression refinement is illustrated in Fig. 6. Note that this approach refers to the classic robust linear regression method, whose implementation is publicly accessible as documented in [44].

The iterative process is essential for several reasons. First, it allows continuous refinement of the calibration constant and data quality. The method progressively eliminates pixel data points that deviate significantly from the expected distribution by iteratively recalculating the residual control interval and removing outliers. This iterative refinement is particularly critical in complex ground surface areas where outliers may persist even after initial robust regression. Second, the iterative approach ensures that anomalies have less influence on the final results, since each iteration strengthens the robustness of the model by focusing on high-quality pixel data points. Finally, this process guarantees that the final radiometric cross-calibration constants are derived from a dataset that is as clean and representative as possible, thereby enhancing the overall accuracy and reliability of the radiometric cross-calibration.

a) Robust regression: It introduces a more robust loss function to reduce the influence of outliers on the traditional OLS method. This method first estimates the parameters using the OLS method, that is [45]

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \cdots + \hat{\beta}_m x_m + \varepsilon \tag{16}$$

where $\hat{y}$ is the predicted value, represents the backscattering coefficient in this article, $\hat{\beta}_0, \hat{\beta}_1, \ldots, \hat{\beta}_m$ are the coefficients estimated by OLS, and ε represents residuals (i.e., $y - \hat{y}$).

To enhance robustness, the standard deviation of the residuals $\hat{s}$ is calculated via median absolute deviation (MAD), since the median has better robustness than the mean when outliers exist for complex ground surface areas. MAD is equal to the median of the absolute value of the new data obtained by subtracting the median from the residuals ε and defined as

$$\text{MAD} = \text{med}\left|\varepsilon_i - \text{med}_j\left(\varepsilon_j\right)\right| \tag{17}$$

where med(·) denotes the median operator. $\hat{s}$ is then scaled to align with the standard normal distribution

$$\hat{s} = \frac{\text{MAD}}{0.6745} \tag{18}$$

where 0.6745 is a correction factor corresponding to the 50% coverage interval of the standard normal distribution (i.e., $\Phi^{-1}(0.75) \approx 0.6745$) [46].

Subsequently, for the convenience of subsequent operations, residuals are standardized to account for leverage effects

$$r_i = \frac{\varepsilon_i}{\hat{s} \cdot \sqrt{1 - h_i}} \tag{19}$$

where h_i is the leverage value of the ith observation, derived from the diagonal elements of the hat matrix $H = X(X^{\mathrm{T}}X)^{-1}X^{\mathrm{T}}$ [47]. High leverage points ($h_i \to 1$) can disproportionately influence the regression results.

Unlike OLS, robust regression minimizes the weighted residual sum

$$\sum_{i=1}^{n} w_i \left(y_i - \sum_{j=1}^{m} x_{ij}\beta_j \right)^2 = \min \tag{20}$$

where w_i is a weight function. This study adopts Tukey's biweight weighting scheme

$$w_i = \begin{cases} 0, & |r_i| > 1 \\ \left(1 - r_i^2\right)^2, & |r_i| \le 1. \end{cases} \tag{21}$$

The weight function assigns weights to each stable pixel point, establishes a weighted least squares estimation model, and continuously refines the weight coefficient through iterations. Due to large residuals, abnormal pixel points are assigned a lower weight, reducing the interference of outliers during the calculation of the calibration constant and effectively enhancing the accuracy of radiometric calibration.

b) Residual control interval calculation and outlier removal: In linear regression for the calibration constant, outlier pixel data points refer mainly to data points that do not belong to the same distribution as the main pixel points, and their

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:27:37 UTC from IEEE Xplore. Restrictions apply.

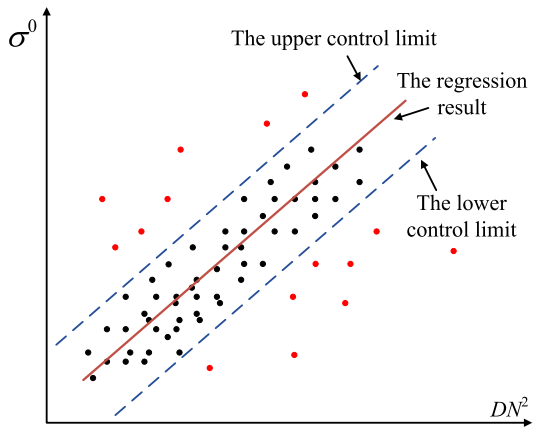


Fig. 7. Residual control interval diagram.

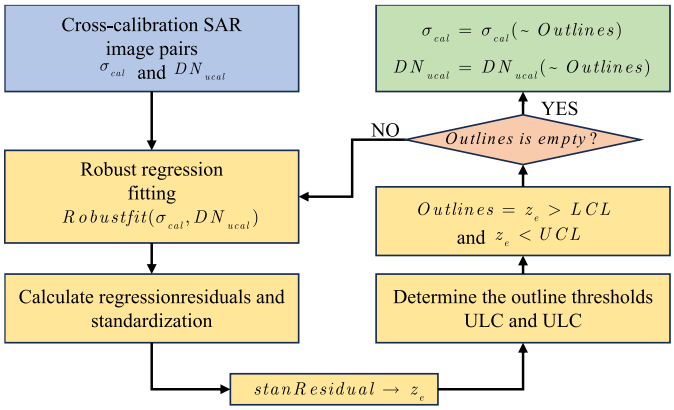


Fig. 8. Algorithm procedure of single-temporal calibration constant calculation based on IRROR.

existence hurts the calculation result. Therefore, based on robust regression, this study further adopts the idea of outlier removal and introduces the standardized residual to measure the quality of the pixel data points. The formula is as follows:

$$z_{\varepsilon_i} = \frac{\varepsilon_i}{s_\varepsilon} \tag{22}$$

where z_{ε_i} is the standardized residual of the pixel data point. $\varepsilon_i = y_i - \hat{y}_i$ is the raw residual, where y_i is the actual value of the ith pixel data point and $\hat{y}_i$ is the predicted value of the ith pixel data point. s_ε is the standard deviation of the residual of the pixel data point

$$s_\varepsilon = \sqrt{\frac{\sum_{i=1}^{n} \varepsilon_i^2}{n-p}} \tag{23}$$

where n is the number of pixel data points and p is the number of model parameters (including the intercept).

After obtaining the robust regression-based calculation result, the standardized residual array of the pixel data points and the upper and lower limits of the residual control are calculated using the formulas, respectively,

$$R = \left[z_{\varepsilon_1}, z_{\varepsilon_2}, z_{\varepsilon_3} \cdots z_{\varepsilon_n}\right] \tag{24}$$

$$\text{UCL} = \mu_R + \phi \cdot \sigma_R \tag{25}$$

$$\text{LCL} = \mu_R - \phi \cdot \sigma_R \tag{26}$$

where R is the standardized residual array of pixel data points, UCL and LCL are the upper and lower limits of the residual control, μ_R is the mean value of the standardized residuals, σ_R is the standard deviation of the standardized residuals, and ϕ is a threshold. We set this value to 2 to ensure that most normal pixel points are retained. Then, taking [LCL, UCL] as the residual control interval, the pixel data points outside this interval are considered outliers, that is, the red points in Fig. 7, which are all eliminated.

The overall algorithmic process for robust regression is shown in Fig. 8: In each iteration, the residuals for each data point are first estimated using OLS. Then, the weighted parameters for the robust regression fit are determined using (17)–(21), and the robust fit is performed using these weighted parameters. Finally, outliers are removed by calculating residual control intervals, and the selected data points are used as input for the next iteration. This iterative process continues until no further outliers are identified, which serves as the convergence criterion.

D. Step 3: Optimization of Time-Series Radiometric Cross-Calibration Constants Based on Kalman Filter

The single-temporal radiometric cross-calibration results exhibit a certain degree of fluctuation, which does not correspond to the actual change in the SAR system's radiometric performance. The actual on-orbit operation state of the SAR sensor is a continuous and gradually evolving process, and the alteration in its radiometric performance is slow and continuous. The radiometric calibration constants at different temporal points are correlated, but existing methods do not consider the correlation of the calibration results at each temporal point.

Although temporal averaging can offer a straightforward way to alleviate fluctuations in the radiometric calibration constants, applying this method essentially disregards temporal variations in the radiometric characteristics of the SAR system.

To address the issues above, we propose a time-series radiometric cross-calibration constant optimization method based on the Kalman filter. The flowchart is illustrated in Fig. 9. After SAR time-series radiometric cross-calibration is implemented, the Kalman filter method is used to optimize the time-series calibration results at each temporal point sequentially, and the optimization results of the calibration constants are promptly updated to the time-series calibration constant curve. The above process is repeated until the entire time-series calibration constant curve is optimized.

1) Kalman Filter: The Kalman filter uses recursive Bayesian estimation to iteratively refine system state predictions through optimal fusion of prior estimates and current measurements [48]. This predictor-corrector algorithm operates through sequential prediction and update phases, where each temporal point combines previous state estimates with new observations to compute statistically optimal solutions under minimum-variance criteria.

A linear stochastic differential system can be characterized by

$$X(t) = A \cdot X(t-1) + B \cdot U(t) + W(t) \tag{27}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:27:37 UTC from IEEE Xplore. Restrictions apply.

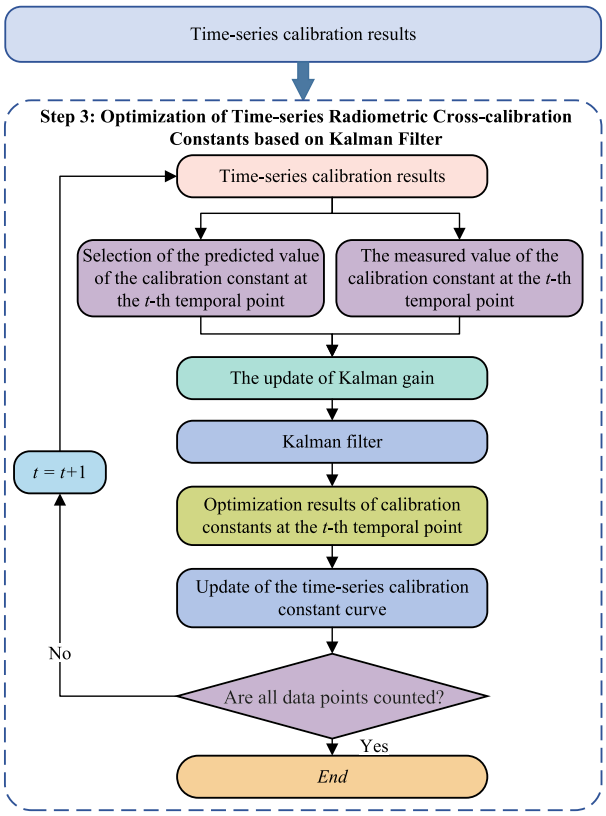


Fig. 9. Flowchart for the optimization of time-series radiometric cross-calibration constants based on Kalman filter.

where $X(t)$ represents the system state at the tth temporal point, $U(t)$ denotes the control quantity of the system at the tth temporal point, which is generally ignored in practical applications, and A and B are the system parameters that define the state transition and control input effects, respectively. Process noise ($W(t)$) follows a zero-mean Gaussian distribution with a covariance matrix (Q), which represents the uncertainty associated with the process noise.

The measurement model is given by

$$Z(t) = H_r \cdot X(t) + V(t) \tag{28}$$

where $Z(t)$ is the measurement value observed at the tth temporal point, H_r is the measurement matrix, and $V(t)$ is the measurement noise with covariance R, which represents the uncertainty associated with the measurement noise.

To facilitate the subsequent derivation of formulas and before delving into the detailed exposition of the Kalman filter algorithm, here we define several key notations: $X(t|t)$ denotes the optimal estimate of the state at time t. $e(t)$ represents the estimation error at time t, defined as the discrepancy between the true state and its estimate. $P(t)$ signifies the mean square error of the state at time t, quantifying the uncertainty associated with the state estimate. $X(t|t-1)$ indicates the optimal prediction of the state $X(t)$ at time t based on information up to time $t-1$ and the state transition model. $e(t|t-1)$ signifies the prediction error at time t, arising from the prediction made using the state at time $t-1$, denoted as $X(t-1|t-1)$ in conjunction with the state transition equation. $P(t|t-1)$, refers to the covariance matrix at time t, given the state prediction based on information up to time $t-1$. It encapsulates the uncertainty related to the predicted state. $X(t)$ stands for the true state at time t.

The algorithm operates sequentially, using the state estimate from the previous time step to predict the current state, and the relationship is expressed as follows:

$$X(t|t-1) = A \cdot X(t-1|t-1) \tag{29}$$

where $X(t|t-1)$ is the prediction result of the current state, and $X(t-1|t-1)$ is the optimal result of the previous state. Subsequently, the prediction error at the tth temporal point can be derived as follows:

$$e(t|t-1) = X(t) - X(t|t-1) = A \cdot e(t-1) + W(t-1). \tag{30}$$

Then, the covariance $P(t|t-1)$ of the prediction result at the current temporal point is

$$\begin{aligned} P(t|t-1) &= E\left(e(t|t-1) \cdot e(t|t-1)\right)^{\mathrm{T}} \\ &= A \cdot P(t-1|t-1) \cdot A^{\mathrm{T}} + Q \end{aligned} \tag{31}$$

where $P(t|t-1)$ is the covariance corresponding to $X(t|t-1)$, and $P(t-1|t-1)$ is the covariance corresponding to $X(t-1|t-1)$. Equations (29) and (31) are the predictions of the system. Combining the predicted value and the observed value of the current state, the optimized estimation value $X(t|t)$ of the current state (that is, the tth temporal point) is obtained as

$$X(t|t) = X(t|t-1) + K(t) \cdot (Z(t) - H \cdot X(t|t-1)) \tag{32}$$

where $K(t)$ denotes the Kalman gain, which can be calculated based on the estimation error $e(t)$ at the tth temporal point

$$\begin{aligned} e(t) &= X(t) - X(t|t) \\ &= X(t) - [X(t|t) + K(t)(Z(t) - HX(t|t-1))] \\ &= e(t|t-1) - K(t)(HX(t) + V(t) - HX(t|t-1)) \\ &= (I - K(t)H)\, e(t|t-1) - K(t)V(t). \end{aligned} \tag{33}$$

Subsequently, the estimated covariance at the tth temporal point can be calculated as follows:

$$\begin{aligned} P(t) &= E\left(e(t)\, e(t)^{\mathrm{T}}\right) \\ &= (I - K(t)H)\, P(t|t-1)\, (I - K(t)H)^{\mathrm{T}} + K(t)RK(t)^{\mathrm{T}}. \end{aligned} \tag{34}$$

The estimated covariance under optimal conditions is minimized when the derivative of $P(t)$ is set to zero with respect to $K(t)$. Consequently, the Kalman gain $K(t)$ can be determined as follows:

$$K(t) = P(t|t-1) \cdot H^{\mathrm{T}} / \left(H \cdot P(t|t-1) \cdot H^{\mathrm{T}} + R\right). \tag{35}$$

The covariance corresponding to the tth state is

$$P(t|t) = (I - K(t) \cdot H) \cdot P(t|t-1) \tag{36}$$

where I is the identity matrix. When the system enters the $(t+1)$th state, $P(t|t)$ becomes $P(t-1|t-1)$ in (32), and thus the algorithm can operate recursively.

This recursive process allows the Kalman filter to efficiently update state estimates in real-time as new measurements become available without reprocessing all previous data.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:27:37 UTC from IEEE Xplore. Restrictions apply.

2) Time-Series Calibration Constants Optimization Based on Kalman Filter: This study adapts the idea of the Kalman filter to optimize time-series radiometric cross-calibration constants through multitemporal calibration constant fusion. The core principle involves reconciling two calibration constants at each temporal point: a predicted value (the prediction of the current temporal point's calibration constant based on the previous temporal points' calibration constants) and a measured value (the measurement value of the current temporal point's calibration constant calculated based on actual data). Both of these values are erroneous. The Kalman gain is calculated based on the errors, and the gain represents the weights of the two values, thereby combining them to obtain the optimal estimated calibration constant for the current temporal point.

Before applying the Kalman filter optimization, four essential parameters must be selected and adjusted for each temporal point according to well-defined criteria to ensure the effectiveness of the subsequent optimization process [49] as follows.

1) *Predicted Calibration Constant ($k_{p,t}$):* Determined as the most reliable radiometric calibration constant within a specified time window preceding the tth epoch. This choice is motivated by the fact that the orbital state of a spaceborne SAR sensor evolves gradually over time.
2) *Prediction Uncertainty (Q_t):* The prediction uncertainty reflects the error associated with the calibration constant at the previous temporal point. Modern SAR missions such as Sentinel-1 typically exhibit slowly varying in-orbit behavior, with radiometric stability better than 0.48 dB per year. Although long-term drift remains small, short-term fluctuations may arise due to thermal noise or minor system variations. Consequently, the initial value of Q_t is conservatively set to 0.5 dB to accommodate potential stepwise variations.
3) *Measured Calibration Constant ($k_{m,t}$):* Obtained from the cross-calibration calculation at the temporal point t.
4) *Measurement Uncertainty (R_t):* Assigned based on the radiometric calibration accuracy of contemporary SAR satellites. For example, since the radiometric accuracy of Sentinel-1 is generally better than 1 dB, an initial value of $R_t = 1$ dB is adopted.

Then, according to the Kalman filtering procedure and (32), the optimal calibration constant at the tth temporal point is given by

$$\hat{k}_t = k_{p,t} + K_t \times \left(k_{p,t} - k_{m,t}\right) \tag{37}$$

where K_t denotes the Kalman gain, computed as

$$K_t = \frac{Q_t^2}{Q_t^2 + R_t^2}. \tag{38}$$

The uncertainty of the optimal value at the tth temporal point is

$$P_t = \sqrt{(1 - K_t)\, Q_t^2}. \tag{39}$$

Upon obtaining the optimized calibration constant at the tth temporal point, the corresponding time-series result is promptly corrected. The uncertainty of this optimal value is then calculated to determine the Kalman gain, which optimizes the $(t+1)$th calibration constant. This iteration continues until all time-series results are optimized (see Fig. 10 for details). Note that the algorithm is derived from improvements to the classic Kalman filter, with its implementation built on the open-source framework documented in [50].

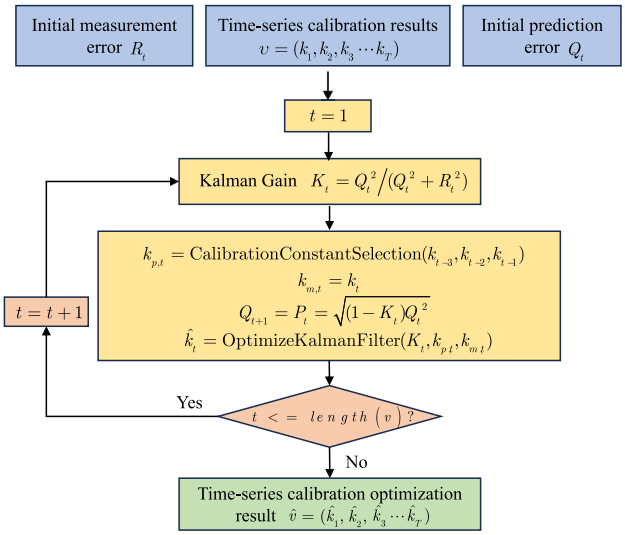


Fig. 10. Algorithm procedure for the optimization of time-series radiometric cross-calibration constants based on Kalman filter.

It is essential to distinguish the dynamic nature of these parameters to ensure the stability of time-series optimization. While the measurement uncertainty R_t is maintained as a constant value derived from the nominal radiometric specifications of the sensor, the prediction uncertainty Q_t employs an adaptive adjustment mechanism. As illustrated in the recursive workflow in Fig. 10, the prediction error for each subsequent temporal point Q_{t+1} is not a fixed heuristic, but is instead directly derived from the posterior covariance of the current state P_t according to the relationship $Q_{t+1} = P_t = ((1 - K_t)Q_t^2)^{1/2}$. This iterative update ensures that the Kalman gain K_t dynamically modulates the weighting between the predicted state and the new observation based on the mathematical confidence of the preceding estimate. By integrating this adaptive feedback loop, the method effectively accommodates potential stepwise variations in sensor performance while suppressing transient measurement noise.

E. Step 4: Construction and Evaluation of SAR Sensor Attenuation Model

Following time-series radiometric cross-calibration at individual temporal points, we establish SAR sensor degradation patterns through the OLS method. The flowchart is illustrated in Fig. 11. The calibration constants are modeled as a linear function of days since launch (DSL)

$$k\,(\text{DSL}) = \alpha \cdot \text{DSL} + k_0 \tag{40}$$

where α is the daily attenuation rate, k_0 is the initial calibration constant, and DSL is the days since satellite launch. Since the value of the daily attenuation rate is small and is difficult to

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:27:37 UTC from IEEE Xplore. Restrictions apply.

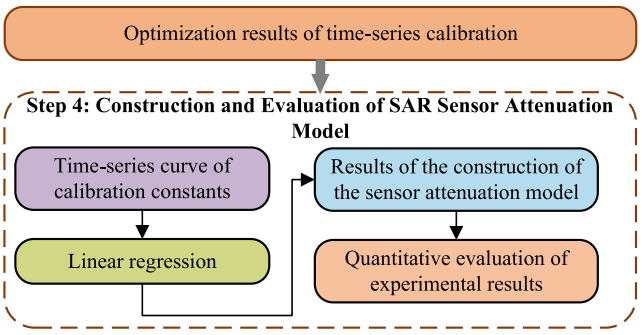


Fig. 11. Flowchart of SAR sensor attenuation model construction and evaluation.

compare and calculate, it is converted to the annual attenuation rate AAR, and the formula is as follows:

$$\mathrm{AAR} = \alpha \cdot 365. \tag{41}$$

The attenuation model clearly demonstrates the drift of the sensor's radiometric performance, thereby enabling an accurate assessment of the overall reliability of the sensor. Moreover, in practical application scenarios, when faced with certain special situations, such as data loss in some periods or errors caused by external factors, it can reasonably predict and fill in the missing calibration constants or effectively interpolate and correct data points with significant errors. Meanwhile, for multisource analysis tasks, the attenuation model can calibrate and normalize data from different sensors.

III. Experiment

In this section, experiments are carried out using the Sentinel-1A/B public dataset to quantitatively evaluate the effectiveness and accuracy of the proposed method.

A. Experimental Area

This study focuses mainly on the types of calibration reference targets of urban areas. According to research by Yang et al. [35], the temporal scattering stability of urban targets is comparable to that of tropical rainforests. Compared with distributed targets such as deserts and bare land, urban targets provide higher backscattering coefficient values, which helps reduce background noise interference. In addition, urban areas often include various targets, such as buildings, water bodies, and vegetation. The distribution ranges of the backscattering coefficient values of these targets are different, and there are often data outliers. This can more effectively verify the effectiveness of the method presented in this article.

The areas selected for the experiment are Lima, Peru, and Los Angeles, USA. The distribution of buildings in these areas is relatively neat and regular, showing a certain regularity. In addition, the average annual precipitation is low, and the influence of external factors on the value of the backscattering coefficient is small. These characteristics make the scattering characteristics of urban targets significant and relatively stable, meeting the standards of calibration targets for radiometric cross-calibration.

B. Experimental Data and Parameters

To validate the proposed method, we use Sentinel-1 data from the European Space Agency's Copernicus program. This series comprises two C-band satellites: Sentinel-1A, launched in April 2014, and Sentinel-1B, launched in April 2016. These satellites offer single-polarization and dual-polarization modes with a 12-day revisit cycle, enabling the continuous acquisition of time-series SAR images essential for our radiometric cross-calibration experiments. For this study, focus is placed on the interferometric wide (IW) mode, widely adopted for land observations because of its balanced spatial resolution and coverage. Specifically, Level-1 ground range detected (GRD) products in IW mode are used.

In the calibration target area screening experiment, time-series SAR images of the same region in different months within a year are obtained. These images have the same orbital directions and polarizations, with closely matching incidence angles. The imaging interval is limited to no more than 60 days to capture the variations in the target scattering characteristics. After preprocessing, the backscattering coefficient images are derived to analyze the spatial histogram characteristics and the temporal scattering stability of the targets.

In the radiometric cross-calibration experiment, SAR image pairs with similar incidence angles and close imaging times are obtained to minimize the impact of the different SAR imaging geometries and the changes of the calibration target on the radiometric cross-calibration results. The radiometric cross-calibration is completed using the backscattering coefficients of the calibrated SAR and the DN of the uncalibrated SAR. The remaining areas in the SAR images are used to verify the radiometric calibration accuracy.

To ensure the validity and reliability of the above experiments, the experimental data are preprocessed by the SNAP and Sentinel Toolboxes software provided by the European Space Agency, including orbit correction, radiometric calibration, terrain correction, etc.

The key parameters used in the methodology are explicitly specified as follows: area_num = 9; block_rows = 50; block_cols = 50; residual control interval $\phi = 2$; prediction error $Q = 0.5$; and measurement error $R = 1$.

Among these, area_num, block_rows, and block_cols are parameters utilized in the extraction of stable targets via the sliding window median method. The residual control interval ϕ serves as a critical parameter in the IRROR algorithm. In addition, the prediction error Q and the measurement error R are parameters specific to the Kalman filter algorithm.

C. Experimental Result Evaluation Metrics

Quantitative assessments are conducted from two perspectives to evaluate the experimental results: single-temporal and time-series radiometric cross-calibration results.

1) Single-Temporal Cross-Calibration Accuracy: The calibration accuracy at each temporal point is quantitatively assessed using the root-mean-square error (RMSE). The RMSE at the tth temporal point is defined as

$$\mathrm{RMSE}_t = \sqrt{\frac{\sum_{i=1}^{m} \left(\sigma_{t,i} - \hat{\sigma}_{t,i}\right)^2}{m}} \tag{42}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:27:37 UTC from IEEE Xplore. Restrictions apply.

where m is the number of data points used for the accuracy calculation at the tth temporal point, $\sigma_{t,i}$ is the calculated backscattering coefficient value obtained by the calibration constant based on the proposed method, and $\hat{\sigma}_{t,i}$ is the true backscattering coefficient value calculated by the official calibration constants. Note that while the official constants serve as the operational standard, they represent a static or low-frequency baseline. Therefore, in this context, a lower RMSE signifies a higher consistency with the standard operational products, validating the proposed method's capability to reproduce official calibration quality using low-cost natural targets.

2) Time-Series Radiometric Cross-Calibration Optimization Performance: To evaluate the optimization performance of the time-series calibration constants, the following quantitative metrics are used: the average calibration constant ($\mathrm{AVG_k}$), the standard deviation of the time-series calibration constants ($\mathrm{STD_k}$), and the AAR.

The average calibration constant is computed as follows:

$$\mathrm{AVG}_k = \frac{1}{T}\sum_{t=1}^{\mathrm{T}} k_t \tag{43}$$

where k_t is the radiometric calibration constant at the tth temporal point, and T is the total number of temporal points. This metric provides an overall measure of the calibration consistency across the time series.

The standard deviation quantifies the stability of the time-series radiometric cross-calibration constants

$$\mathrm{STD}_k = \sqrt{\frac{\sum_{t=1}^{\mathrm{T}} (k_t - \mathrm{AVG}_k)^2}{T-1}}. \tag{44}$$

A lower value of $\mathrm{STD_k}$ indicates higher stability in the calibration constants over time.

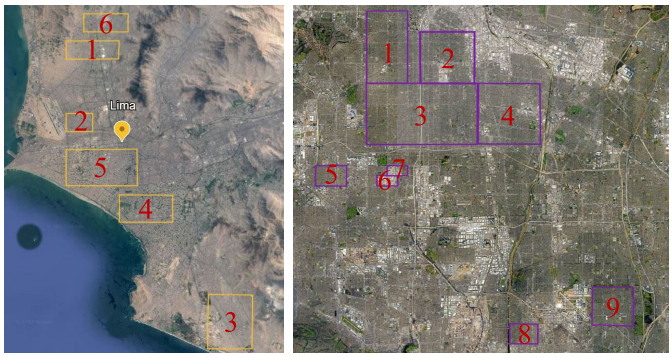
In addition, the calculation formula for the AAR is shown in (41), which reflects the changes in the radiometric performance of the SAR sensor.

D. Screening Results of the Calibration Target Area

This section presents the screening process and results of the target area for radiometric cross-calibration, considering both the satellite revisit frequency, the spatial histogram characteristics, and the temporal scattering stability.

From a theoretical point of view, the proposed spatiotemporal filtering–regression–modeling framework is designed to be fundamentally independent of specific land-cover types or SAR sensor configurations. The method operates on general statistical and physical principles, such as temporal consistency, distributional stability, and robust regression, which ensure its applicability to diverse surface conditions. Lima and Los Angeles are used only as examples for demonstrating and validating the workflow and do not restrict the applicability of the method. Although other natural invariant targets are not included in this study, the proposed method is general and can be applied to regions such as deserts or salt flats.

1) Screening Result by Satellite Revisit Frequency Analysis: The revisit cycle of the Sentinel-1 satellites is 12 days. Some images may exhibit quality issues, so the DQF is set to 0.95. According to statistics, from August 15, 2019, to December 2, 2021 (a total of 840 days), 69 pairs of images are available for radiometric cross-calibration in Los Angeles. It can be known through (1) that this number meets the threshold requirement. Therefore, the urban area data of Los Angeles is selected for the subsequent time-series radiometric cross-calibration. In contrast, the urban area data for Lima does not meet the required threshold for the number of high-quality image pairs. Therefore, Lima is only used for the subsequent single-temporal radiometric cross-calibration experiment, where fewer image pairs are sufficient for validation purposes.



(a) (b)

Fig. 12. Location of target subareas. (a) Lima area. (b) Los Angeles area.

2) Screening Result by Target Spatial Histogram Characteristics: Following the preprocessing steps, including orbit correction and radiometric calibration, backscattering coefficient images are derived for Los Angeles and Lima. Subsequently, the histogram of the target SAR image is analyzed to identify suitable regions based on specific criteria. The selection process focuses on areas with uniform backscattering coefficients, free of abnormal bright spots, and exhibiting a unimodal histogram distribution. In addition, the histograms must have approximately symmetric left and right sides, along with a short-tailed distribution. These criteria ensure the stability and reliability of the selected target areas for subsequent radiometric cross-calibration experiments. Ultimately, six target subareas in Lima and nine target subareas in Los Angeles meet the selection requirements. Their respective spatial distribution and distribution histograms are presented in Figs. 12 and 13, visually demonstrating the consistency and quality of the selected regions.

3) Screening Result by Target Temporal Scattering Characteristics: Subsequently, time-series SAR data from Sentinel-1A over one year, covering different months, is collected. The focus is on the above target subareas in Lima and Los Angeles. Each dataset is acquired under consistent conditions, including similar incidence angles, identical orbital directions (ascending or descending), matching polarization settings, and overlapping imaging areas. Furthermore, the imaging interval between consecutive acquisitions is constrained to no more than 60 days to ensure temporal consistency. Detailed information on the datasets is provided in Tables I (Lima) and II (Los Angeles). After preprocessing, such as orbit correction, radiometric calibration, speckle filtering, and geocoding, the time-series

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:27:37 UTC from IEEE Xplore. Restrictions apply.

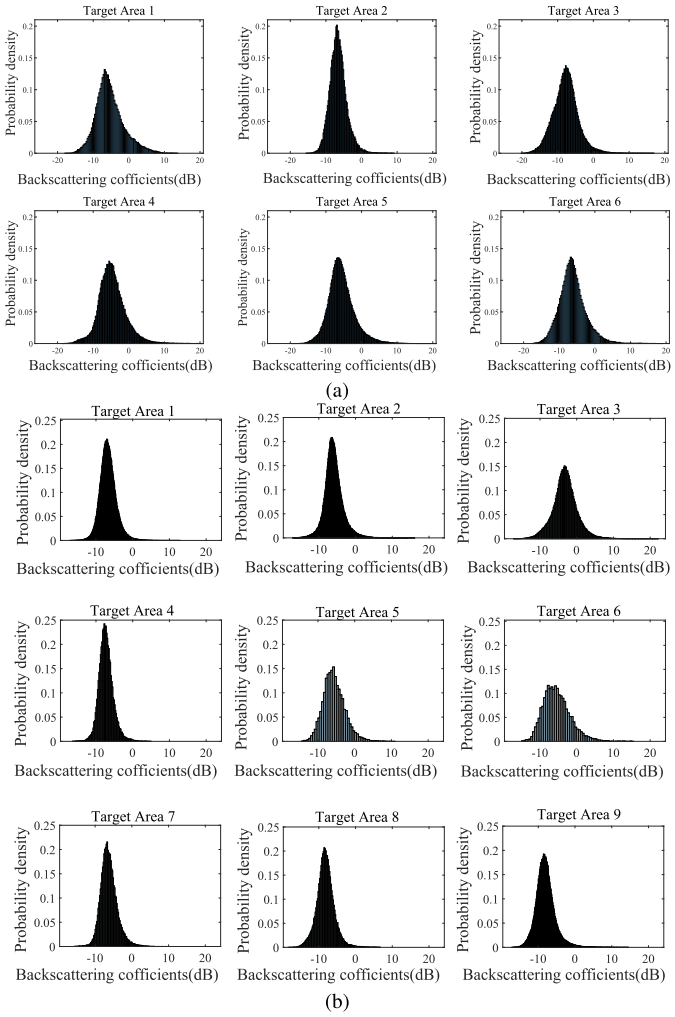


Fig. 13. Histogram of the target subareas distribution. (a) Lima area. (b) Los Angeles area.

TABLE I
ACQUISITION DETAILS OF SENTINEL-1A OVER LIMA

Area	Satellite	Acquisition Date	Orbit Direction	Incidence Angle
Lima	Sentinel-1A	2022/02/26	Ascending	37.3°
Lima	Sentinel-1A	2022/03/10	Ascending	37.3°
Lima	Sentinel-1A	2022/04/03	Ascending	37.3°
Lima	Sentinel-1A	2022/05/09	Ascending	37.3°
Lima	Sentinel-1A	2022/06/02	Ascending	37.3°
Lima	Sentinel-1A	2022/07/08	Ascending	37.3°
Lima	Sentinel-1A	2022/08/13	Ascending	37.3°
Lima	Sentinel-1A	2022/09/18	Ascending	37.3°
Lima	Sentinel-1A	2022/10/24	Ascending	37.3°

backscattering coefficient images of these candidate subareas are obtained, respectively.

To ensure reliable results for radiometric cross-calibration, subareas with temporal scattering stability below 0.5 dB are identified as scattering stable areas and selected as the target regions according to (7). Their corresponding temporal scattering stabilities and location information are summarized in Tables III (for Lima) and IV (for Los Angeles).

TABLE II
ACQUISITION DETAILS OF SENTINEL-1A OVER LOS ANGELES

Area	Satellite	Acquisition Date	Orbit Direction	Incidence Angle
Los Angeles	Sentinel-1A	2020/01/12	Descending	38.4°
Los Angeles	Sentinel-1A	2020/02/17	Descending	38.4°
Los Angeles	Sentinel-1A	2020/04/05	Descending	38.4°
Los Angeles	Sentinel-1A	2020/05/23	Descending	38.4°
Los Angeles	Sentinel-1A	2020/07/10	Descending	38.4°
Los Angeles	Sentinel-1A	2020/08/27	Descending	38.4°
Los Angeles	Sentinel-1A	2020/10/02	Descending	38.4°
Los Angeles	Sentinel-1A	2020/11/19	Descending	38.4°
Los Angeles	Sentinel-1A	2020/12/25	Descending	38.4°

TABLE III
LOCATION INFORMATION AND TIME-SERIES SCATTERING STABILITY OF TARGET SUBAREAS IN LIMA

Region	Longitude and Latitude	Temporal Stability (dB)
1	11°57′30″S,77°04′30″W	0.4240
2	12°01′30″S,77°05′15″W	0.3561
3	12°12′30″S,76°56′44″W	0.3028
4	12°06′14″S,77°01′21″W	0.4709
5	12°04′00″S,77°04′00″W	0.4183
6	11°56′00″S,77°03′45″W	0.3216

TABLE IV
LOCATION INFORMATION AND TIME-SERIES SCATTERING STABILITY OF TARGET SUBAREAS IN LOS ANGELES

Region	Longitude and Latitude	Temporal Stability (dB)
1	33°28′43″N, 118°18′34″W	0.3814
2	33°58′19″N, 118°15′32″W	0.3424
3	33°57′11″N, 118°12′16″W	0.3065
4	33°54′32″N, 118°21′32″W	0.4989
5	33°54′45″N, 118°18′02″W	0.4983
6	33°54′20″N, 118°18′32″W	0.4716
7	33°47′46″N, 118°11′33″W	0.4496
8	33°49′01″N, 118°07′01″W	0.3766
9	33°47′31″N, 117°57′38″W	0.4260

The experimental results demonstrate that the six subareas screened in Lima and the nine subareas screened in Los Angeles all meet the essential requirements for radiometric cross-calibration. Specifically, these subareas exhibit unimodal, symmetrical, and short-tailed distributions in their histograms. Moreover, these subareas exhibit excellent temporal scattering stability.

4) Acquisition of Time-Series SAR Image Pairs: As mentioned above, the frequency of satellite overlapping transits in Lima does not meet the requirements; only the frequency in Los Angeles reaches the results calculated by (1). As a result, Los Angeles's subareas in Fig. 12(b) are finally selected for the subsequent time-series radiometric cross-calibration experiment. The image pairs available for radiometric cross-calibration in this area from August 2019 to December 2021 contain 69 groups, with an average of approximately 12 days per group. Based on the location information of the nine subareas in Los Angeles, the calibrated and uncalibrated SAR image pairs are obtained, temporal point by temporal point.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:27:37 UTC from IEEE Xplore. Restrictions apply.

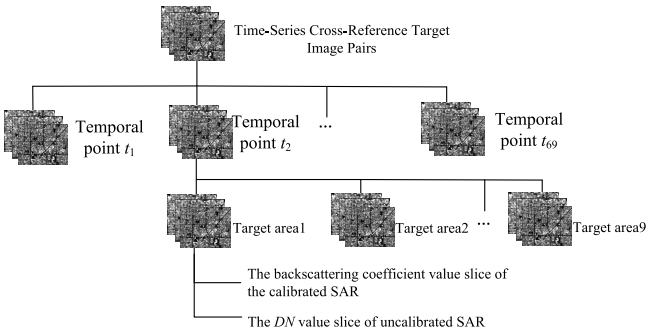


Fig. 14. Diagram of time-series SAR image pairs.

TABLE V
EXPERIMENTAL DATA INFORMATION

Experiment Type	Data	Date	Angle	Satellite
Cross-Calibration of the Same Satellite	Uncalibrated	2022/11/29	37.3°	Sentinel-1A
	Calibrated	2022/12/11	37.3°	Sentinel-1A
Cross-Calibration of Different Satellite	Uncalibrated	2021/09/09	38.4°	Sentinel-1B
	Calibrated	2021/09/15	38.4°	Sentinel-1A

The image pair structure is shown in Fig. 14. Each image pair belongs to a single-temporal point and contains the backscattering coefficient image slice of the calibrated SAR and the DN image slice of the uncalibrated SAR for the nine subareas.

E. Calculation Results of Single-Temporal Radiometric Cross-Calibration Constants

In this section, taking the Lima area and the Los Angeles area as references, the effects of the proposed method for radiometric cross-calibration of the same satellite (Sentinel-1A and Sentinel-1A) and different satellites (Sentinel-1A and Sentinel-1B) are tested, respectively. To reduce the influence of background noise on the results, image data in the VV polarization is selected for the experiment. The information of the experimental data is presented in Table V.

Firstly, the calibrated and uncalibrated SAR images are preprocessed. The backscattering coefficient value of the calibrated SAR data and the DN value of the uncalibrated SAR data are extracted using the window median method. Then, the relationship between $\sigma^0_{\text{caledSAR}}$ and $\text{DN}_{\text{uncalSAR}}$ is obtained using four regression schemes: OLS, Huber regression, random sample consensus (RANSAC), and the IRROR algorithm. The fit results for the same-satellite case (Lima) and the cross-satellite case (Los Angeles) are shown in Fig. 15(a) and (b), respectively.

For clarity, only the effective sample points are plotted, whereas potential outliers are automatically down-weighted or removed by the robust estimators. Across both test sites, the proposed IRROR consistently achieves the highest calibration accuracy among the four regression schemes.

In addition, to quantitatively evaluate the accuracy of the proposed IRROR method, the radiometric calibration of the uncalibrated image is performed using the calibration equation above, and the experimental results of the backscattering

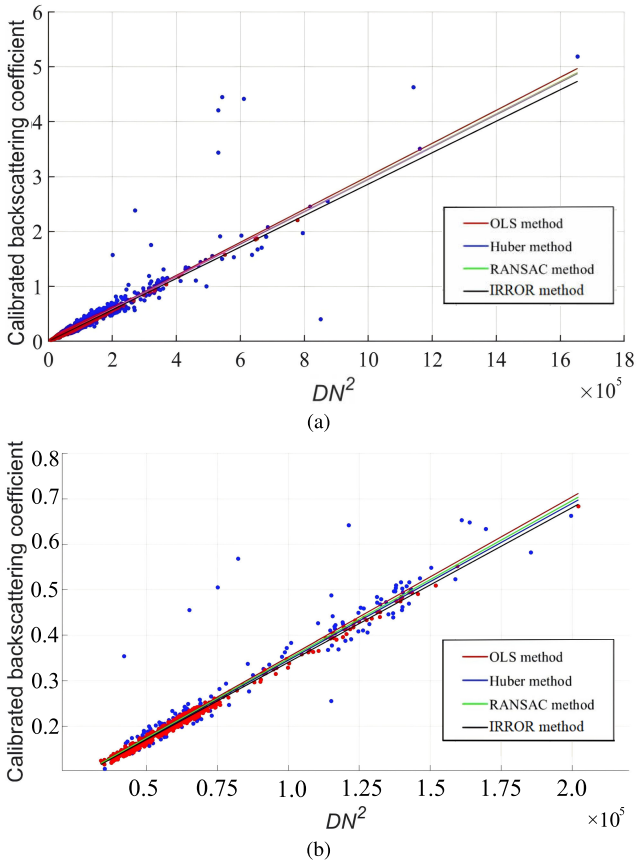


Fig. 15. Single-temporal radiometric cross-calibration results. (a) Same satellite at Lima. (b) Different satellites in Los Angeles.

TABLE VI
CALIBRATION CONSTANT (k) AND RMSE COMPARISON AMONG OLS, RANSAC, HUBER, AND IRROR FOR LIMA AND LOS ANGELES

Area	Method	k ($\times 10^{-6}$)	RMSE (dB)
Lima (same satellite)	OLS	3.0038	0.283
	RANSAC	2.6908	0.223
	Huber	2.9578	0.217
	IRROR	2.8620	0.074
Los Angeles (different satellites)	OLS	3.5213	0.797
	RANSAC	3.4805	0.769
	Huber	3.4505	0.736
	IRROR	3.4034	0.643

coefficient are obtained. Subsequently, the official calibration constants (radiometric calibration method based on artificial calibration targets) are applied to process the data, resulting in the official backscattering coefficient. Finally, the radiometric calibration accuracy, represented by RMSE, is calculated using (42). The calibration accuracy results obtained from the experiment are summarized in Table VI.

In the radiometric cross-calibration experiment for the same satellite, the conventional OLS approach achieved an RMSE of 0.283 dB. The proposed IRROR method effectively mitigates the influence of outliers by adaptively down-weighting anomalous samples, thereby reducing RMSE to 0.074 dB, an improvement of about 0.21 dB. Compared with other robust regression methods such as RANSAC and Huber, IRROR

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:27:37 UTC from IEEE Xplore. Restrictions apply.

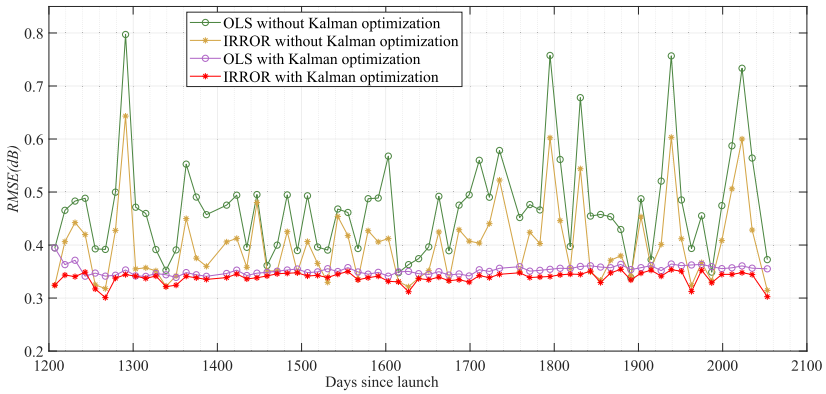


Fig. 16. RMSE comparison of the four experiments: performance of OLS and IRROR with and without Kalman Optimization.

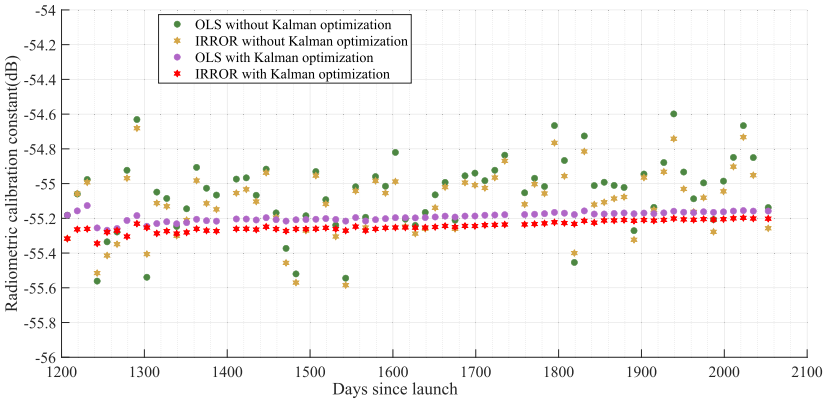


Fig. 17. Time-series radiometric cross-calibration constants of the four experiments: performance of OLS and IRROR with and without Kalman optimization.

more accurately preserves the true radiometric relationship by suppressing extreme deviations without discarding valid observations.

In radiometric cross-calibration between different satellites, where systematic biases and observation inconsistencies are more pronounced, OLS yielded an RMSE of 0.797 dB, whereas IRROR reduced it to 0.643 dB, achieving an improvement of approximately 0.15 dB. This shows that the robustness of the method to heterogeneity and outliers directly contributes to enhanced cross-satellite consistency, highlighting its potential for standardized SAR radiometric calibration.

F. Optimization Results of Time-Series Radiometric Cross-Calibration Constants

Four experiments are designed to evaluate the effectiveness of the proposed time-series optimization methods as follows.

1) *Experiment 1:* Traditional OLS with independent time-series calibration.
2) *Experiment 2:* The proposed IRROR with independent time-series calibration.
3) *Experiment 3:* Traditional OLS optimized by the Kalman filter.
4) *Experiment 4:* The proposed IRROR optimized by the Kalman filter.

The experiments are conducted using 69 SAR image pairs acquired from August 15, 2019, to December 2, 2021. The results of the four experiments are summarized in Table VII. The time-series calibration accuracy and the results are shown in Figs. 16 and 17.

1) Comparison Between IRROR and OLS: As demonstrated in Fig. 16, the proposed IRROR method outperforms the traditional OLS method in terms of calibration accuracy in 69 temporal points. Specifically, more than 95% of the calibration constants obtained using IRROR exhibit lower RMSE values than those obtained using OLS, with all RMSE values below 0.65 dB. This indicates that the proposed IRROR method also performs well when applied to time-series radiometric calibration of a large number of high-frequency image pairs, showing good reliability and robustness.

Fig. 17 presents the time-series calibration constants derived from both methods. The average calibration constant (AVG_k) obtained using IRROR is −55.11 dB, with a standard deviation (STD_k) of 0.1916 dB, an improvement of 0.0503 dB over the OLS result (0.2419 dB). This highlights the improved stability of the IRROR method.

2) Kalman Filter Optimization Effects: The Kalman filter method is applied to optimize the time-series radiometric cross-calibration constants to obtain the optimized time-series calibration constant curve. The results are presented in Figs. 16 and 17.

For the OLS method, the Kalman filter improves the accuracy of the calibration at 67 out of 69 temporal points, with all RMSE values below 0.40 dB. Only at the 56th and 64th

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:27:37 UTC from IEEE Xplore. Restrictions apply.

TABLE VII
EXPERIMENTAL RESULTS

No.	Group of Comparison	Average of $RMSE$ (dB)	Average of Δ_{RMSE} (dB)	Max of Δ_{RMSE} (dB)	Number of Δ_{RMSE} (dB)	STD_K (dB)
1	OLS without Kalman optimization	0.472	0.076	0.156	68	0.2419
2	IRROR with Kalman optimization	0.396				0.1916
3	OLS without Kalman optimization	0.472	0.12	0.444	67	0.2419
4	OLS with Kalman optimization	0.352				0.0300
5	IRROR without Kalman optimization	0.396	0.075	0.300	68	0.1916
6	IRROR with Kalman optimization	0.321				0.0269

temporal points is the accuracy slightly lower than that of the unoptimized OLS-based calibration. The optimized average calibration constant (AVG_k) is −55.19 dB, and the standard deviation (STD_k) is 0.03 dB, which is 0.2119 dB better than the unoptimized OLS method (0.2419 dB).

For the IRROR method, the Kalman filter improves the calibration accuracy at 68 out of 69 temporal points, with all RMSE values below 0.36 dB. Only the calibration accuracy of the 27th temporal point is slightly lower than the individual IRROR-based calibration without optimization. The optimized average calibration constant (AVG_k) is −55.24 dB, and the standard deviation (STD_k) is 0.0269 dB, a marked improvement of 0.1647 dB over the unoptimized IRROR method (0.1916 dB).

3) Sensor Attenuation Model: Finally, based on the above time-series calibration constants, the sensor attenuation model is constructed using the OLS method, and the specific result is as follows.

For OLS without Kalman optimization

$$k\,(\mathrm{DSL}) = 2.838 \times 10^{-4} \times \mathrm{DSL} - 55.518. \tag{45}$$

The initial calibration constant is −55.518 dB and the AAR is 0.104 dB.

For IRROR without Kalman optimization

$$k\,(\mathrm{DSL}) = 2.376 \times 10^{-4} \times \mathrm{DSL} - 55.499. \tag{46}$$

The initial calibration constant is −55.499 dB, and the AAR is 0.086 dB according to (41).

For OLS with Kalman optimization

$$k\,(\mathrm{DSL}) = 7.620 \times 10^{-5} \times \mathrm{DSL} - 55.315. \tag{47}$$

The initial calibration constant is −55.315 dB and the AAR is 0.028 dB.

For IRROR with Kalman optimization

$$k\,(\mathrm{DSL}) = 1.087 \times 10^{-4} \times \mathrm{DSL} - 55.422. \tag{48}$$

The initial calibration constant is −55.422 dB and the AAR is 0.0397 dB.

These results indicate that the proposed methods effectively suppress fluctuations in the calibration constant curve, providing a more accurate representation of the SAR sensor's on-orbit state.

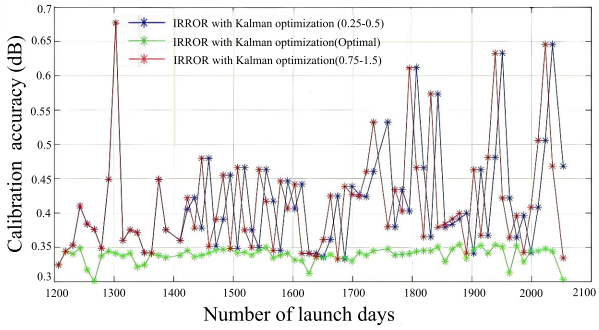


Fig. 18. SAR time-series radiometric cross-calibration accuracy under different Kalman filter parameter settings.

G. Sensitivity Analysis of Kalman Filter Parameters

To further investigate the robustness of the proposed Kalman filter optimization, we conducted a sensitivity analysis with respect to the initial prediction and measurement error settings. Fig. 18 illustrates the time-series radiometric cross-calibration accuracy under three distinct parameter configurations. The green curve represents the calibration accuracy obtained with the optimal parameter settings of $Q_t = 0.5$ and $R_t = 1$ adopted in this study. In contrast, the blue and red curves correspond to nonoptimal parameter combinations of $(Q_t, R_t) = (0.25, 0.5)$ and $(0.75, 1.5)$, respectively.

The rationale for selecting these specific nonoptimal sets lies in characterizing the response of the filter to two divergent deviation scenarios relative to the established baseline. The parameter combination of $(Q_t, R_t) = (0.25, 0.5)$ simulates a condition of underestimated uncertainty with tight constraints, which assesses the susceptibility of the system to noise when overconfidence is placed in both model predictions and measurements. In contrast, the set $(Q_t, R_t) = (0.75, 1.5)$ represents an overestimation of error statistics through loose constraints, testing whether the filter excessively suppresses valid signal variations or introduces instability due to assumed high noise levels.

As can be observed in Fig. 18, the optimal configuration produces the most stable performance with a calibration accuracy consistently maintained within the range of 0.3–0.35 dB. In contrast, the nonoptimal parameter choices result in significant instability, as both the blue and red curves exhibit pronounced oscillations with local peaks exceeding 0.6 dB.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:27:37 UTC from IEEE Xplore. Restrictions apply.

These deviations, characterized by frequent abrupt jumps in accuracy across the time series, highlight the susceptibility of the Kalman filter to inappropriate parameter initialization. This experiment confirms that careful selection of initial prediction and measurement error values is essential to ensure the effectiveness of the Kalman filter. The optimal parameters not only suppress the influence of outliers on the calibration constants but also lead to smoother accuracy curves, thereby validating the parameter setting strategy adopted in this study.

H. Computational Complexity and Efficiency Analysis

The computational complexity of the proposed method is approximately $O(N \log N)$, where N represents the total number of pixel samples. A primary factor contributing to this efficiency is the rapid convergence of the IRROR algorithm, which consistently requires fewer than five iterations to reach a stable solution. As established in the methodology, the convergence criterion is strictly governed by the outlier detection process described in Section II-C3b, whereby the iterative loop proceeds recursively until the dataset reaches a state where no further outliers are identified. This deterministic stopping condition ensures both statistical rigor and the reproducibility of the results while maintaining minimal computational overhead. Furthermore, the Kalman filter optimization operates with $O(T)$ complexity relative to the number of temporal points T, ensuring excellent scalability for extended observation periods.

In terms of practical performance, the comprehensive processing of 69 temporal points across 9 calibration subareas on a 13th Gen Intel Core i5-13400 CPU with 16-GB RAM is completed in approximately 30 min. These performance metrics demonstrate that the proposed method achieves a favorable tradeoff between computational expenditure and radiometric accuracy, rendering it highly suitable for large-scale and long-term SAR time-series applications.

IV. Discussion

The proposed SAR time-series radiometric cross-calibration method effectively addresses limitations of conventional methods. By integrating multicriterion target screening (satellite revisit frequency, spatial histogram characteristics, and temporal scattering stability), the method enables reliable calibration using naturally distributed targets, overcoming the high costs and limited scalability of artificial calibration targets. The IRROR method effectively mitigates the interference of outliers, improving the single-temporal calibration accuracy by 0.076 dB compared to the OLS method. In addition, Kalman filter optimization improves temporal stability, reducing the standard deviation of calibration constants by 89% in experiments, thereby providing a robust solution for long-term monitoring of sensor performance.

Beyond its practical effectiveness, the proposed method embodies several key theoretical innovations that fundamentally advance SAR time-series radiometric cross-calibration. First, the multicriterion target-screening mechanism establishes a quantitative link among satellite revisit frequency, spatial histogram morphology (characterized by skewness and kurtosis), and temporal scattering stability (STD < 0.5 dB). This coupling addresses the long-standing decoupling issue between spatial and temporal criteria in conventional calibration, providing a unified theoretical basis for selecting targets that are both spatially homogeneous and temporally stable. Second, IRROR introduces a residual-controlled iterative loop that adaptively suppresses anomalous samples based on standardized residuals and control limits. This innovation extends classical robust statistics to heterogeneous SAR scenes, effectively stabilizing parameter estimation under non-Gaussian noise conditions. Third, Kalman filter optimization with physical constraints incorporates a sensor attenuation model into the prediction equation and parameterizes process and measurement errors using empirical radiometric stability data (e.g., <0.48 dB per year for Sentinel-1). These modifications bridge statistical modeling with physical sensor behavior, ensuring that temporal smoothing preserves the physical interpretability of radiometric drift.

The thresholds used in target screening are not arbitrary. They are derived from statistical distribution principles, SAR scattering physics, and radiometric calibration constraints. This integration ensures that the selected targets maintain both spatial uniformity and temporal stability, satisfying the core requirements of time-series radiometric cross-calibration. The innovation of this study also lies in the coupling of robust estimation theory and state-space modeling, which provides a unified spatiotemporal optimization paradigm. By embedding robust regression within a dynamic Kalman filtering process, the method not only suppresses measurement noise but also models sensor drift, achieving consistent and physically interpretable calibration across temporal sequences. This theoretically grounded integration extends the conceptual boundary of SAR radiometric calibration and provides a generalizable approach for future multisensor and multitemporal applications.

Despite these strengths, the proposed method has certain limitations. Its dependence on adequate satellite overlap restricts its applicability to regions with sufficient acquisition frequency, as demonstrated by Lima, where the temporal sampling density does not meet the required threshold. The screening criteria for urban targets (skewness < 0.1 and kurtosis < 0) can also exclude valid, yet complex areas, while environmental variability, especially heavy precipitation, can reduce temporal stability (STD > 0.5 dB) in humid regions. Furthermore, the current validation focuses solely on VV polarization in the IW mode, leaving other configurations (e.g., VH polarization, Stripmap, or Spotlight modes) unexplored. The other configurations may pose additional challenges to histogram-based screening and time-series continuity.

V. Conclusion

This study establishes a unified SAR time-series radiometric cross-calibration method that integrates target screening, robust regression, outlier control, and dynamic Kalman filtering, effectively improving radiometric accuracy, temporal stability, and calibration reliability, demonstrating its capability for long-term monitoring of SAR sensor performance. The results also confirm that urban regions can serve as practical calibration fields once stable scatterers are selected through

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:27:37 UTC from IEEE Xplore. Restrictions apply.

time-series analysis, as the integrated robust regression mechanism inherently suppresses the influence of unstable pixels and ensures reliable long-term calibration even in complex environments. From a theoretical perspective, the proposed method is not restricted to specific sensors or land-cover types and thus possesses inherent universality. Future work will conduct extended validation using the RADARSAT-2 (24-day revisit cycle) and Gaofen-3 (full polarization) datasets. These experiments will assess the adaptability of the cross-sensors between different orbital configurations, polarization modes, and spatial resolutions. Although only Sentinel-1 data were analyzed in this study, the proposed method can be directly extended to natural invariant targets such as deserts and salt flats without methodological modification. The Sentinel-1 results presented herein therefore serve as a proof-of-concept demonstration of a broader, transferable framework capable of supporting next-generation multisensor radiometric harmonization in global Earth observation missions.

REFERENCES

[1] A. Moreira, P. Prats-Iraola, M. Younis, G. Krieger, I. Hajnsek, and K. P. Papathanassiou, "A tutorial on synthetic aperture radar," *IEEE Geosci. Remote Sens. Mag.*, vol. 1, no. 1, pp. 6–43, Mar. 2013.

[2] S. Zhang, Q. Xu, H. Wang, Y. Kang, and X. Li, "Automatic waterline extraction and topographic mapping of tidal flats from SAR images based on deep learning," *Geophys. Res. Lett.*, vol. 49, no. 2, p. 2021, Jan. 2022.

[3] D. Amitrano et al., "Earth environmental monitoring using multi-temporal synthetic aperture radar: A critical review of selected applications," *Remote Sens.*, vol. 13, no. 4, p. 604, Feb. 2021.

[4] F. Gao et al., "SAR target incremental recognition based on features with strong separability," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5202813.

[5] S. Wangchuk, T. Bolch, and B. A. Robson, "Monitoring glacial lake outburst flood susceptibility using Sentinel-1 SAR data, Google Earth engine, and persistent scatterer interferometry," *Remote Sens. Environ.*, vol. 271, Mar. 2022, Art. no. 112910.

[6] A. Freeman, "SAR calibration: An overview," *IEEE Trans. Geosci. Remote Sens.*, vol. 30, no. 6, pp. 1107–1121, Nov. 1992.

[7] Y. Hu, Y. Nie, Z. Liu, G. Wu, and W. Fan, "Improving the potential of coniferous forest aboveground biomass estimation by integrating C{-} and L-band SAR data with feature selection and non-parametric model," *Remote Sens.*, vol. 15, no. 17, p. 4194, Aug. 2023.

[8] Q. Yin, J. Li, Y. Zhou, D. Xiang, and F. Zhang, "Adaptive weighted learning for vegetation contribution in soil moisture inversion using PolSAR data," *Int. J. Remote Sens.*, vol. 43, no. 9, pp. 3190–3215, Jun. 2022.

[9] Q. Yin et al., "Coherence matrix power model for scattering variation representation in multi-temporal PolSAR crop classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 9797–9810, 2024.

[10] C. R. N. Rao and J. Chen, "Revised post-launch calibration of the visible and near-infrared channels of the Advanced Very High Resolution Radiometer (AVHRR) on the NOAA-14 spacecraft," *Int. J. Remote Sens.*, vol. 20, no. 18, pp. 3485–3491, Jan. 1999.

[11] H. Cosnefroy, M. Leroy, and X. Briottet, "Selection and characterization of Saharan and Arabian desert sites for the calibration of optical satellite sensors," *Remote Sens. Environ.*, vol. 58, no. 1, pp. 101–114, Oct. 1996.

[12] A. K. Heidinger, J. T. Sullivan, and C. R. N. Rao, "Calibration of visible and near-infrared channels of the NOAA-12 AVHRR using time series of observations over deserts," *Int. J. Remote Sens.*, vol. 24, no. 18, pp. 3635–3649, Jan. 2003.

[13] W. Wei, Y. Zhang, M. Zhang, C. Zhao, X. Li, and X. Zheng, "Multisite high-frequency radiometric calibration of GF-1 wide field of view," *Acta Photonica Sinica*, vol. 47, no. 2, pp. 154–161, 2018.

[14] F. Cabot, O. Hagolle, and P. Henry, "Relative and multitemporal calibration of AVHRR, SeaWiFS, and VEGETATION using POLDER characterization of desert sites," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2000, pp. 2188–2190.

[15] Y. Wang et al., "Time-series cross-radiometric calibration and validation of GF-6/WFV using multi-site," *Remote Sens.*, vol. 16, no. 7, p. 1287, Apr. 2024.

[16] L. Liu, T. Shi, H. Gao, X. Zhang, Q. Han, and X. Hu, "Long-term cross calibration of HJ-1A CCD1 and Terra MODIS reflective solar bands," *Sci. Rep.*, vol. 11, no. 1, Apr. 2021, Art. no. 7386.

[17] X. Dou et al., "Time-series calibration and verification of GF1-WFV based on Kalman filtering," *Acta Optica Sinica*, vol. 42, no. 12, p. 8, 2022.

[18] M. Pinheiro et al., "Sentinel-1 instruments status and product performance update for 2022," in *Proc. 14th Eur. Conf. Synth. Aperture Radar*, France, Jul. 2022, pp. 1–5.

[19] A. Bojarski et al., "TanDEM-X long-term system performance after 10 years of operation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 2522–2534, 2021.

[20] Q. Huang et al., "Quick quality assessment and radiometric calibration of C-SAR/01 satellite using flexible automatic corner reflector," *Remote Sens.*, vol. 15, no. 1, p. 104, Dec. 2022.

[21] J. Reimann et al., "The DLR SAR calibration center," in *Proc. IEEE 5th Asia–Pacific Conf. Synth. Aperture Radar (APSAR)*, Sep. 2015, pp. 169–173.

[22] P. Rizzoli, B. Braeutigam, and M. Zink, "TanDEM-X large-scale study of tropical rainforests for spaceborne SAR calibration in X-band," in *Proc. 10th Eur. Conf. Synth. Aperture Radar*, Jun. 2014, pp. 1–4.

[23] X. Chen, Y. Zhou, F. Ma, Q. Yin, and F. Zhang, "Optimized selection of pseudo-invariant calibration sites for radiometric cross-calibration of spaceborne SAR sensors," in *Proc. SAR Big Data Era (BIGSARDATA)*, Sep. 2023, pp. 1–4.

[24] L. Shi et al., "Stability analysis of radiometric and polarimetric system errors of LT-1A satellite's fully polarized SAR: A case study of tropical rainforest," *J. Radars*, vol. 13, no. R24102, p. 1, 2024.

[25] S. Shangguan, X. Qiu, and K. Fu, "Research on a special hyper-pixel for SAR radiometric monitoring," *Remote Sens.*, vol. 15, no. 8, p. 2175, Apr. 2023.

[26] Y. Zhou, L. Zhuang, J. Duan, F. Zhang, and W. Hong, "Synthetic aperture radar radiometric cross calibration based on distributed targets," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 9599–9612, 2022.

[27] M. Deng et al., "Spaceborne SAR radiometric cross-calibration considering typical scattering effects in building areas," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5225611.

[28] A. Freeman, "Radiometric calibration of SAR image data," *Int. Arch. Photogramm. Remote Sens.*, vol. 29, p. 212, Aug. 1993.

[29] S. Gao, Y. Li, T. Bai, Y. Zhang, and X. Zheng, "Uncertainty analysis in cross-calibration and improved calculation of calibration coefficients," *Chin. Opt.*, vol. 13, no. 3, pp. 568–576, 2020.

[30] Y. Zhou, B. Yang, Q. Yin, F. Ma, and F. Zhang, "Improved SAR radiometric cross-calibration method based on scene-driven incidence angle difference correction and weighted regression," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5224216.

[31] Y. Zhou, X. Chen, Q. Yin, F. Ma, and F. Zhang, "SAR radiometric cross-calibration based on multiple pseudoinvariant calibration sites with extensive backscattering coefficient range," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 4836–4849, 2025.

[32] L. Sun et al., "On-orbit response variation analysis of FY-3 MERSI reflective solar bands based on Dunhuang site calibration," *Spectrosc. Spect. Anal.*, vol. 32, no. 7, Sep. 2012, Art. no. 186977.

[33] G. Qiu, X. Li, W. Wei, Q. Zhang, and X. Zheng, "Experiment and analysis of on-orbit radiometric calibration for remote sensors based on in-site automated observation technology," *Acta Optica Sinica*, vol. 36, no. 7, 2016, Art. no. 0701001.

[34] E. Makas and A. R. Aslan, "Spaceborne SAR system design considerations: Minimizing satellite size and mass, system parameter trade-offs, and optimization," *Appl. Sci.*, vol. 14, no. 21, p. 9661, Oct. 2024.

[35] J. Yang, X. Qiu, C. Ding, B. Lei, and X. Lu, "Analyzing and extracting stable feature of target backscattering for C-band SAR," *J. Univ. Chin. Acad. Sci.*, vol. 36, no. 1, p. 10, 2019.

[36] J. Yang, X. Qiu, C. Ding, and B. Lei, "Identification of stable backscattering features, suitable for maintaining absolute synthetic aperture radar (SAR) radiometric calibration of Sentinel-1," *Remote Sens.*, vol. 10, no. 7, p. 1010, Jun. 2018.

[37] F. Ma, X. Sun, F. Zhang, Y. Zhou, and H.-C. Li, "What catch your attention in SAR images: Saliency detection based on soft-superpixel lacunarity cue," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5200817.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:27:37 UTC from IEEE Xplore. Restrictions apply.

[38] M. Huang, H. Zhao, and Y. Chen, "Research on SAR image quality evaluation method based on improved Harris hawk optimization algorithm and XGBoost," *Sci. Rep.*, vol. 14, no. 1, Nov. 2024, Art. no. 28364.
[39] R. Chattamvelli and R. Shanmugam, *Descriptive Statistics for Scientists and Engineers: Applications in R*. Cham, Switzerland: Springer, 2023.
[40] G. Hatem, J. Zeidan, M. Goossens, and C. Moreira, "Normality testing methods and the importance of skewness and kurtosis in statistical analysis," *BAU J. Sci. Technol.*, vol. 3, no. 2, 2022.
[41] C. E. Shannon, "A mathematical theory of communication," *Bell Syst. Tech. J.*, vol. 27, no. 4, pp. 623–656, 1948.
[42] M. Hamze et al., "Integration of L-band derived soil roughness into a bare soil moisture retrieval approach from C-band SAR data," *Remote Sens.*, vol. 13, no. 11, p. 2102, May 2021.
[43] M. Skolnik, *Radar Handbook*. New York, NY, USA: McGraw-Hill, 1990.
[44] J. Poerschmann. (2021). *Methods of Robust Linear Regression*. [Online]. Available: https://github.com/jakobap/Methods-of-Robust-Linear-Regression
[45] S. Wang et al., "Robust regression soil moisture retrieval method based on GNSS-IR multi-satellite fusion," *Bull. Surv. Mapp.*, no. 3, pp. 69–74, 2024.
[46] F. R. Hampel, "Beyond location parameters: Robust concepts and methods," *Bull. Int. Statist. Inst.*, vol. 46, no. 1, pp. 375–382, 1975.
[47] P. E. Johnson, "The hat matrix and regression diagnostics," in *Proc. Joint Stat. Meet.*, 2006, pp. 1–14.
[48] R. E. Kalman, "A new approach to linear filtering and prediction problems," *J. Basic Eng.*, vol. 82, no. 1, pp. 35–45, Mar. 1960.
[49] D. Peng, "Basic principle and application of Kalman filter," *Softw. Guide*, vol. 8, no. 11, pp. 32–34, 2009.
[50] P. Jaiswal. (2018). *Kalman Filter Implementation*. [Online]. Available: https://github.com/PercyJaiswal/KalmanFilter



Yongsheng Zhou (Member, IEEE) received the B.E. degree in communication engineering from Beijing Information Science and Technology University, Beijing, China, in 2005, and the Ph.D. degree in signal and information processing from the Institute of Electronics, Chinese Academy of Sciences, Beijing, in 2010.

He was with the Academy of Opto-Electronics, Chinese Academy of Sciences, from 2010 to 2019. He is currently a Professor of electronic and information engineering with the College of Information Science and Technology, Beijing University of Chemical Technology, Beijing. His research interests include SAR radiometric calibration, SAR target detection and recognition, and remote sensing image processing.



Jing Su (Student Member, IEEE) is currently pursuing the master's degree with the College of Information Science and Technology, Beijing University of Chemical Technology, Beijing, China. She is majoring in information and communication engineering.

Her research interests include SAR image processing and SAR calibration.



Lingling Ma (Member, IEEE) received the M.S. and Ph.D. degrees in geography and geographical information systems from the Chinese Academy of Sciences, Beijing, China, in 2005 and 2008, respectively.

She is currently a Professor with the Aerospace Information Research Institute, Chinese Academy of Sciences. Her research interests include calibration and validation techniques of remote sensing data and products, and in-flight performance assessment of optical sensors.



Qiang Yin (Senior Member, IEEE) received the B.S. degree in electronic and information engineering from Beijing University of Chemical Technology, Beijing, China, in 2004, and the M.S. and Ph.D. degrees in signal and information processing from the Institute of Electronics, Chinese Academy of Sciences, Beijing, in 2008 and 2016, respectively.

From 2008 to 2013, she was a Research Assistant with the Institute of Electronics, Chinese Academy of Sciences. From 2014 to 2015, she was a Research Fellow with the European Space Agency, Rome, Italy. She is currently an Associate Professor with the College of Information Science and Technology, Beijing University of Chemical Technology. Her research interests include polarimetric/polarimetric interferometric synthetic aperture radar (SAR) and deep learning.



Fan Zhang (Senior Member, IEEE) received the B.E. degree in communication engineering from the Civil Aviation University of China, Tianjin, China, in 2002, the M.S. degree in signal and information processing from Beihang University, Beijing, China, in 2005, and the Ph.D. degree in signal and information processing from the Institute of Electronics, Chinese Academy of Sciences, Beijing, in 2008.

From 2013 to 2015, he was a Visiting Scholar working with Prof. Martin D. F. Wong with the Coordinated Science Laboratory, University of Illinois at Urbana–Champaign, Urbana, IL, USA. He is currently a Full Professor of electronic and information engineering with Beijing University of Chemical Technology, Beijing. He has authored or co-authored more than 100 peer-reviewed papers, including IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, *ISPRS Journal of Photogrammetry and Remote Sensing*, CVPR, and ICCV. His research interests include synthetic aperture radar system simulation, SAR image interpretation, and artificial intelligence.

Dr. Zhang is an Editorial Board Member of *Journal of Radars*.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:27:37 UTC from IEEE Xplore. Restrictions apply.